Midterm Project Report

Name: Do Nam Hai
Student ID: 23070846
Email: 23070846@vnu.edu.vn

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Abstract

This project evaluates classical image-processing methods for denoising, stereo reconstruction, and four-image panorama stitching. Mean, Gaussian, and median filters with Laplacian sharpening were compared on synthetic Gaussian and salt-and-pepper noise using PSNR, SSIM, MSE, and runtime. Gaussian filtering achieved the highest selected PSNR for Gaussian noise, while a 3×3 median filter performed best for impulse noise; sharpening gains depended on configuration. On the rectified Venus pair, SGBM provided higher valid-disparity coverage and lower ground-truth disparity error than BM across tested settings, with greater runtime. Reconstructed point clouds use an assumed camera model and therefore show qualitative normalized depth only. For stitching, SIFT, ORB, and AKAZE were evaluated with RANSAC homographies and feather blending; SIFT had the lowest selected pooled reprojection error, while ORB showed greater corner sensitivity. The results reveal trade-offs among fidelity, coverage, speed, and geometric stability.

1 Introduction

Image filtering, stereo reconstruction, and image stitching address three practical problems in computer vision: improving image quality, recovering scene structure, and combining overlapping views. Noise can obscure fine details, uncertain correspondences can distort reconstructed geometry, and inaccurate alignment can produce visible seams in a panorama. Traditional computer vision provides explicit methods for these tasks through local filtering, feature matching, and geometric estimation. Their parameters and intermediate outputs make it possible to examine how processing choices affect both numerical measurements and visible image quality.

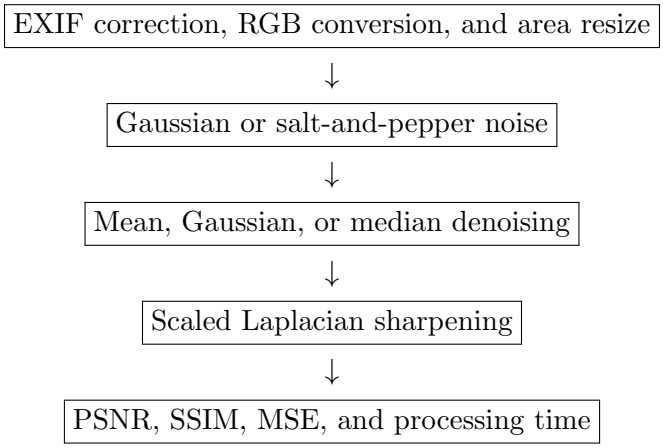
This project investigates these methods in three parts. Part A compares mean, Gaussian, and median filtering under Gaussian and salt-and-pepper noise, followed by Laplacian sharpening to examine the balance between noise removal and detail enhancement. Part B studies stereo correspondences, fundamental matrix estimation, and disparity computation using Block Matching (BM) and Semi-Global Block Matching (SGBM), with attention to the camera assumptions required for 3D reconstruction. Part C compares feature detectors and descriptors in a pipeline that matches, aligns, and blends four overlapping images into a panorama. Each part presents the methodology, implementation and results, and comparative analysis. Quantitative measurements and visual comparisons are used together to assess accuracy, runtime, and failure cases, followed by the overall conclusions and possible improvements.

2 Part A: Image Filtering

2.1 Methodology

Reference image and processing workflow.

The input is `succulent_20231108_171619.jpg`, a photograph containing succulent leaves, fine gravel, a glazed pot, and a broad background. Its encoded dimensions are 4128×3096 pixels. Correcting EXIF orientation 6 produces a 3096×4128 RGB image, which is resized to 768×1024 using OpenCV's `INTER_AREA`. No crop is applied to the reference, and the original JPEG is preserved. The available metadata records the original filename and local path but does not identify the photographer or usage terms; whether the image was self-captured could not be verified. All quality measurements use this resized image before synthetic corruption as the reference.



A separate control sharpens the noisy image directly. Both the denoised and sharpened outputs are evaluated against the same reference.

Noise models.

Gaussian noise consists of independent zero-mean samples added to each RGB element, with standard deviation $\sigma_n \in \{10, 20, 30\}$ in 8-bit intensity units:

$$N_G = \text{round}\big[\text{clip}_{[0,255]}(I + \epsilon)\big], \qquad \epsilon \sim \mathcal{N}(0, \sigma_n^2). \tag{1}$$

Addition uses floating-point values. Clipping and rounding can change the realized noise distribution, so its measured mean and standard deviation are retained in the run manifest.

Salt-and-pepper noise selects round(pHW) distinct spatial pixels. The tested densities are $p \in \{0.01, 0.03, 0.05\}$. Approximately half are set to white and half to black across all channels. The selected fraction and the fraction whose values actually change are recorded separately. Seeds 42, 43, and 44 are fixed; random-stream identity also includes the noise type and level. Every method receives the same corrupted image for each noise condition and seed.

Denoising and sharpening.

For a $k \times k$ neighbourhood Ω_k, the mean filter is

$$D_M(x, y, c) = \frac{1}{k^2} \sum_{(u,v)\in\Omega_k} N(x + u, y + v, c). \tag{2}$$

The Gaussian filter uses a finite normalized kernel:

$$G(u, v) = \frac{\exp[-(u^2 + v^2)/(2\sigma_f^2)]}{\sum_{(a,b)\in\Omega_k} \exp[-(a^2 + b^2)/(2\sigma_f^2)]}, \qquad D_G = G * N. \tag{3}$$

The median filter replaces each channel value with the median of its neighbourhood. These operations use `cv2.blur`, `cv2.GaussianBlur`, and `cv2.medianBlur`, independently per channel. Replicated borders are used throughout, including the median filter's native border handling.[1]

Laplacian enhancement follows denoising in the main pipeline:

$$J = \text{round}\big[\text{clip}_{[0,255]}\left(D - \alpha s_k \Delta_k D\right)\big], \qquad s_k = 2^{4-2k}. \tag{4}$$

The derivative is computed with `cv2.Laplacian` in float32 to preserve its sign. Apertures $k \in \{3, 5, 7\}$ use scales 1/4, 1/64, and 1/1024 to control derivative gain across apertures. The strength is $\alpha \in \{0.1, 0.2, 0.25, 0.5\}$. Sharpening enhances local contrast and is assessed separately from noise removal.[2]

Quality metrics and runtime.

For $N = 3HW$ RGB elements, the reference I and result J are compared using

$$\text{MSE} = \frac{1}{N}\sum_{i=1}^{N}(I_i - J_i)^2, \qquad \text{PSNR} = 10\log_{10}\frac{255^2}{\text{MSE}}. \tag{5}$$

Differences and squares use float64 to avoid unsigned-integer overflow. MSE is reported in squared 8-bit intensity units. Zero MSE gives infinite PSNR. SSIM measures local luminance, contrast, and structure:

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}, \tag{6}$$

where $C_1 = (0.01\cdot255)^2$ and $C_2 = (0.03\cdot255)^2$. The implementation uses `data_range=255`, `channel_axis=-1`, Gaussian window weights with sigma 1.5, and population covariance. Channel scores are averaged. The SSIM window sigma is distinct from the filter and noise sigmas.[3]

Quality is reported as mean ± sample standard deviation over three seeds; PSNR is averaged in dB. Each processed case has two warm-ups and ten timed repetitions per seed. A configuration's runtime is the median of its 30 samples, with interquartile range (IQR) in parentheses. Timing includes processing checks and array conversions, and excludes noise generation, metric computation, disk I/O, and plotting. Denoising–sharpening chains are timed as complete operations. OpenCV uses one thread with OpenCL disabled. No filter runtime is assigned to noisy baselines.

[1] OpenCV filtering API: `https://docs.opencv.org/4.x/d4/d86/group__imgproc__filter.html`.

[2] OpenCV Laplace operator: `https://docs.opencv.org/4.x/d5/db5/tutorial_laplace_operator.html`.

[3] scikit-image quality metrics: `https://scikit-image.org/docs/stable/api/skimage.metrics.html#skimage.metrics.structural_similarity`.

2.2 Implementation and Results

Experimental setup and parameter grid.

The experiment was implemented with OpenCV, NumPy, Pillow, scikit-image, and Matplotlib. Measurements were collected with Python 3.13.14 on Linux x86_64 using NumPy 2.5.4, opencv-python-headless 5.0.0.93, Pillow 13.0.0, scikit-image 0.26.0, and Matplotlib 3.11.2. The accompanying reproducibility package records the input checksum, source snapshot, parameters, environment, and realized noise statistics.

The final grid uses denoising kernels $k \in \{3, 5, 7, 9, 11\}$ and Gaussian filter sigmas $\sigma_f \in \{0.6, 0.8, 1.0, 1.2, 1.4, 1.5, 1.6\}$. Each of the six noise conditions has 45 denoising settings: five mean, five median, and 35 Gaussian combinations. An earlier coarse, refined, and boundary search covered 382 distinct denoiser configurations and 22 Gaussian sigma values across its searches; it was not a full Cartesian product of those values.

For each noise condition and denoiser family, the highest three-seed mean whole-image PSNR selects the input to sharpening; ties prefer smaller kernels. Each condition then has 36 denoising–sharpening settings and six direct-sharpening controls at $\alpha = 0.25$ or 0.5. Including the six noisy baselines, the experiment contains 528 configuration groups, 1,584 seed-level metric rows, and 15,660 timing samples. All 528 seed-42 cases are saved as PNG files along with the reference. The accompanying machine-readable results retain every tested setting, including poorly performing configurations.

Noisy baselines and denoising results.

Table 1 reports the corrupted-reference baselines. Tables 2 and 3 compare the PSNR-selected configuration within each denoiser family. Appendix B reports every configuration in the final grid; the machine-readable results retain the full precision and individual seeds.

Table 1: Noisy baselines. Gaussian levels denote noise standard deviation; impulse levels denote the fraction of selected pixels. Quality is mean ± sample standard deviation over three seeds.

Noise	Level	PSNR (dB)	SSIM	MSE
Gaussian	$\sigma_n = 10$	28.2319 ± 0.0001	0.54723 ± 0.00010	97.70 ± 0.00
Gaussian	$\sigma_n = 20$	22.3294 ± 0.0008	0.28528 ± 0.00013	380.31 ± 0.07
Gaussian	$\sigma_n = 30$	18.9570 ± 0.0035	0.17793 ± 0.00011	826.76 ± 0.67
Salt-and-pepper	1%	25.0383 ± 0.0313	0.74063 ± 0.00112	203.82 ± 1.47
Salt-and-pepper	3%	20.2892 ± 0.0058	0.44027 ± 0.00076	608.35 ± 0.81
Salt-and-pepper	5%	18.0649 ± 0.0108	0.28699 ± 0.00171	1015.29 ± 2.52

Selected denoising–sharpening chains.

Table 4 lists the highest mean PSNR at each noise level, selected from the measured denoised outputs and complete sharpening chains. Direct controls are excluded from this selection. The accompanying results package maps each selected configuration to its seed-42 output image and checksum.

The selected chains improve PSNR over their denoised inputs by 0.1423, 0.0809, and

Table 2: Gaussian-noise denoising comparison. Quality is mean ± sample standard deviation; time is median (IQR), in milliseconds. The filter sigma is σ_f.

Noise level	Filter	PSNR (dB)	SSIM	MSE	Time (ms)
10	G 5×5, $\sigma_f=0.8$	34.2954 ± 0.0021	0.87793 ± 0.00003	24.18 ± 0.01	0.974 (0.294)
10	M 3×3	33.7557 ± 0.0018	0.87724 ± 0.00005	27.38 ± 0.01	0.422 (0.037)
10	Med 3×3	33.5676 ± 0.0021	0.84035 ± 0.00002	28.60 ± 0.01	0.325 (0.050)
20	G 7×7, $\sigma_f=1.2$	31.2319 ± 0.0089	0.82511 ± 0.00014	48.97 ± 0.10	2.456 (0.419)
20	M 5×5	30.4018 ± 0.0064	0.83307 ± 0.00013	59.28 ± 0.09	0.532 (0.041)
20	Med 5×5	30.6926 ± 0.0124	0.79309 ± 0.00037	55.44 ± 0.16	3.226 (0.542)
30	G 9×9, $\sigma_f=1.5$	29.4227 ± 0.0033	0.79417 ± 0.00018	74.27 ± 0.06	3.045 (0.677)
30	M 5×5	28.9585 ± 0.0025	0.75971 ± 0.00012	82.65 ± 0.05	0.540 (0.032)
30	Med 7×7	29.0048 ± 0.0118	0.77319 ± 0.00042	81.77 ± 0.22	60.777 (3.619)

Table 3: Salt-and-pepper denoising comparison. Quality is mean ± sample standard deviation; time is median (IQR), in milliseconds. The filter sigma is σ_f.

Density	Filter	PSNR (dB)	SSIM	MSE	Time (ms)
1%	G 5×5, $\sigma_f=1.0$	32.6387 ± 0.0243	0.87682 ± 0.00011	35.42 ± 0.20	1.149 (0.470)
1%	M 3×3	32.1472 ± 0.0257	0.85450 ± 0.00016	39.66 ± 0.24	0.417 (0.038)
1%	Med 3×3	38.2479 ± 0.0093	0.96514 ± 0.00002	9.73 ± 0.02	0.324 (0.024)
3%	G 7×7, $\sigma_f=1.4$	30.1131 ± 0.0131	0.81756 ± 0.00053	63.35 ± 0.19	2.725 (1.215)
3%	M 5×5	29.5823 ± 0.0088	0.80084 ± 0.00046	71.59 ± 0.14	0.628 (0.908)
3%	Med 3×3	37.8470 ± 0.0330	0.96393 ± 0.00006	10.68 ± 0.08	0.449 (0.347)
5%	G 11×11, $\sigma_f=1.6$	28.7862 ± 0.0185	0.79405 ± 0.00008	85.99 ± 0.37	4.042 (1.001)
5%	M 5×5	28.3379 ± 0.0179	0.74163 ± 0.00012	95.34 ± 0.39	0.683 (0.194)
5%	Med 3×3	37.4485 ± 0.0597	0.96265 ± 0.00003	11.70 ± 0.16	0.437 (0.166)

Table 4: Selected complete chains. G denotes Gaussian noise (level σ_n); SP denotes impulse density. Filter sizes are square; Gaussian settings show kernel width / σ_f. L is the Laplacian aperture. Quality is mean ± sample standard deviation; time is median (IQR).

Noise	Denoiser	$L;\alpha$	PSNR (dB)	SSIM	MSE	Time (ms)
G10	Gauss 5/.8	7; 0.1	34.4378 ± 0.0015	0.87848 ± 0.00002	23.40 ± 0.01	11.814 (1.477)
G20	Gauss 7/1.2	7; 0.1	31.3128 ± 0.0087	0.82536 ± 0.00014	48.06 ± 0.10	12.552 (1.664)
G30	Gauss 9/1.5	7; 0.1	29.4742 ± 0.0035	0.79435 ± 0.00017	73.39 ± 0.06	12.570 (1.028)
SP 1%	Med 3	5; 0.25	38.7671 ± 0.0048	0.96794 ± 0.00002	8.64 ± 0.01	9.217 (0.772)
SP 3%	Med 3	5; 0.25	38.2426 ± 0.0459	0.96655 ± 0.00006	9.75 ± 0.10	12.210 (4.236)
SP 5%	Med 3	5; 0.2	37.7493 ± 0.0626	0.96474 ± 0.00004	10.92 ± 0.16	12.355 (2.826)

0.0515 dB for Gaussian $\sigma_n = 10, 20, 30$, and by 0.5192, 0.3956, and 0.3009 dB for impulse densities 1%, 3%, and 5%. Gaussian-noise improvements are modest relative to the added cost: at $\sigma_n = 20$, the selected denoiser takes 2.456 ms, compared with 12.552 ms for the complete chain. Denoising alone is a reasonable choice when processing speed matters more than this small fidelity gain.

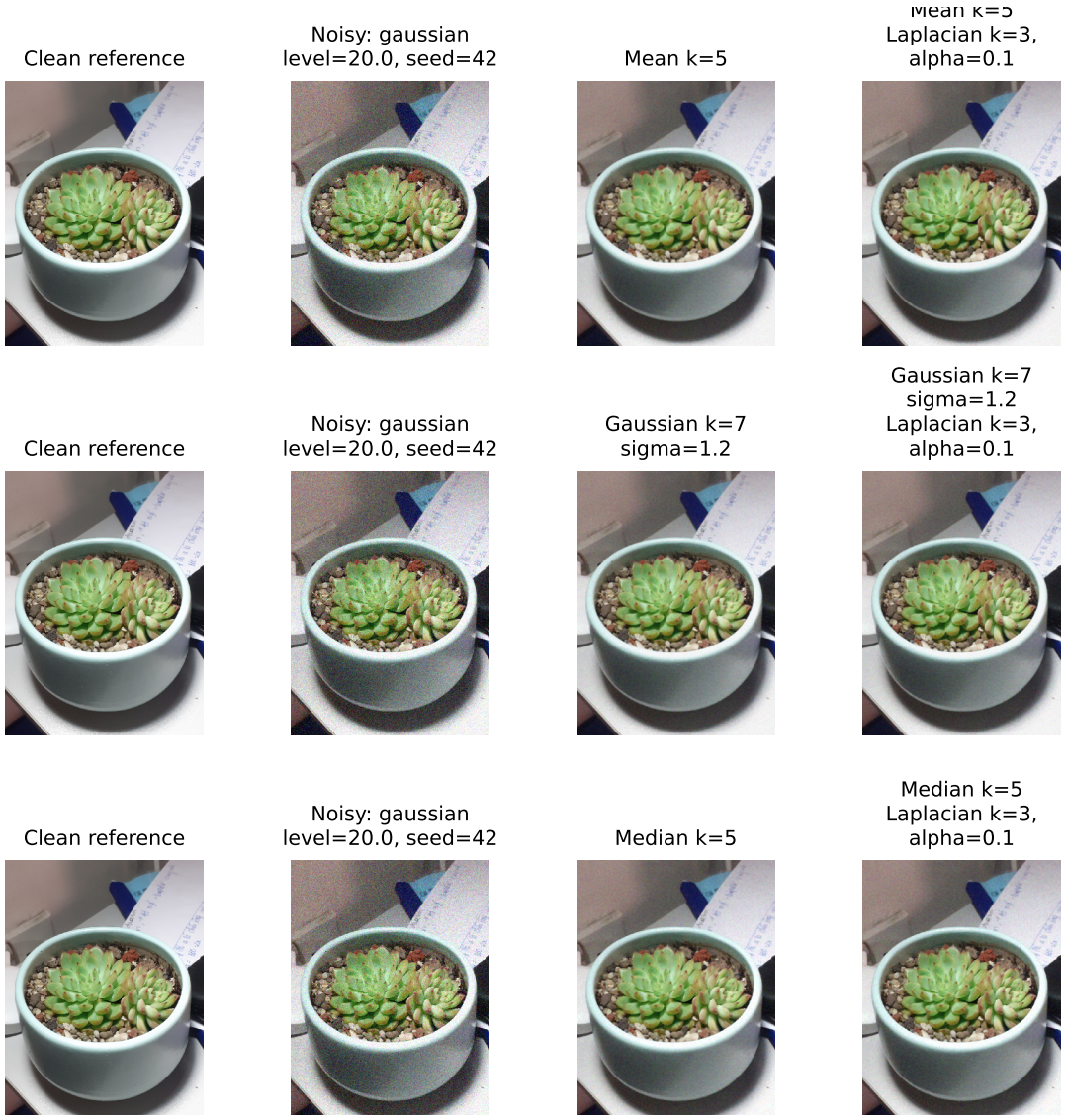


Figure 1: Gaussian noise $\sigma_n = 20$, seed 42. Rows compare the selected mean, Gaussian, and median denoisers. Columns show original, noisy, denoised, and sharpened images. The fourth column uses illustrative Laplacian $k = 3$, $\alpha = 0.1$; the selected chains in Table 4 use separately selected parameters.

2.3 Comparative Analysis

Noise removal and edge preservation.

Gaussian filtering gives the highest selected denoising PSNR at all three Gaussian noise levels. At $\sigma_n = 20$, Gaussian 7×7, $\sigma_f = 1.2$ reaches 31.2319 dB, compared with 30.4018 dB for mean 5×5 and 30.6926 dB for median 5×5. Mean filtering is faster but its equal weighting blurs fine gravel and leaf tips. The PSNR-selected Gaussian setting does not maximize every metric: mean 5×5 gives higher SSIM at this level, 0.83307 versus 0.82511.

Median 3×3 has the highest selected PSNR and SSIM at every tested impulse density. At 3%, it increases PSNR from 20.2892 to 37.8470 dB and SSIM from 0.44027 to 0.96393. Its median runtime is 0.449 ms, compared with 2.725 ms for the selected Gaussian filter. Median filtering rejects sparse extrema without spreading them across the neighbourhood.

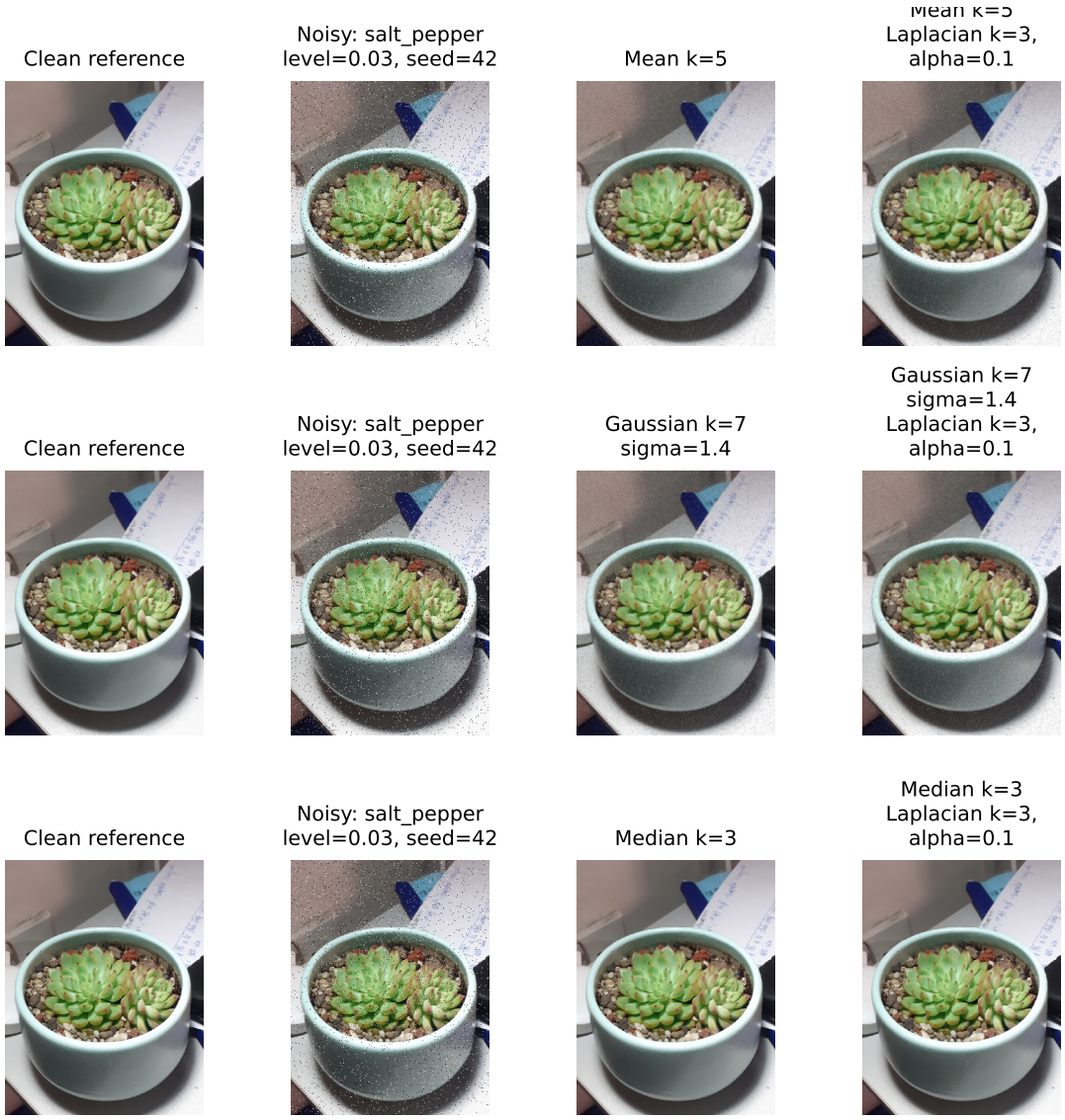


Figure 2: Salt-and-pepper noise $p = 3\%$, seed 42, with the same column order and illustrative sharpening parameters as Figure 1.

Larger median kernels remove small gravel structures and are unnecessary at these tested densities. Runtime also depends strongly on aperture: the selected median 7×7 at Gaussian $\sigma_n = 30$ takes 60.777 ms, compared with 3.045 ms for Gaussian 9×9, $\sigma_f = 1.5$.

Smooth regions, texture, and the optimization objective.

Three fixed diagnostic regions cover the pot surface, leaf boundaries, and gravel (Figure 4). They use the same coordinates for every method and are not alternative experiment inputs. Whole-image metrics remain the primary selection criterion. The pot surface makes residual grain visible, leaf boundaries expose blur and contrast changes, and gravel exposes fine-texture loss.

At Gaussian $\sigma_n = 20$, the highest whole-image denoising SSIM in the final grid is obtained by Gaussian 11×11, $\sigma_f = 1.6$: SSIM 0.86334 and PSNR 30.7340 dB. The PSNR-selected 7×7, $\sigma_f = 1.2$ instead gives SSIM 0.82511 and PSNR 31.2319 dB. Table 5 compares their regional scores for seed 42 only.

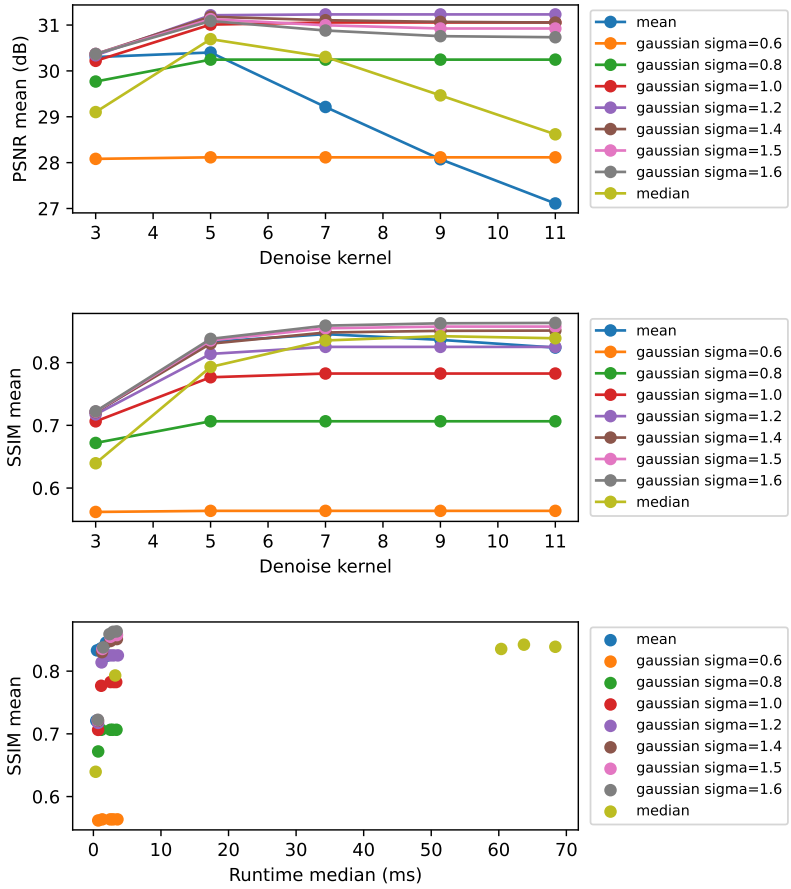


Figure 3: Kernel sensitivity and quality/runtime trade-offs at Gaussian noise $\sigma_n = 20$. Quality averages the three seeds; time is the pooled median.

Figure 4: Reference image and the three fixed diagnostic regions. The experiment reference remains uncropped.

Table 5: Regional quality for seed 42 only. PSNR choice: Gaussian 7×7, $\sigma_f = 1.2$; SSIM choice: Gaussian 11×11, $\sigma_f = 1.6$. These are not three-seed means.

Selection	Region	PSNR (dB)	SSIM
PSNR	Pot surface	34.4065	0.80543
PSNR	Leaf edges	30.2437	0.83638
PSNR	Gravel	27.4009	0.83000
SSIM	Pot surface	36.6267	0.89023
SSIM	Leaf edges	29.3627	0.83990
SSIM	Gravel	25.8209	0.78849

Broader Gaussian smoothing raises pot-surface PSNR from 34.4065 to 36.6267 dB, while reducing gravel PSNR from 27.4009 to 25.8209 dB and gravel SSIM from 0.83000 to 0.78849. Better smooth-region scores can therefore accompany damaged texture. Leaf SSIM is slightly higher with broader smoothing, so neither setting dominates every regional metric. The PSNR choice retains more gravel fidelity, although residual Gaussian grain remains visible.

Figure 5: Mean 3×3 versus 11×11 in the gravel region at Gaussian $\sigma_n = 20$, seed 42. The broader kernel merges fine structures and softens nearby boundaries; the notebook compares all five kernel sizes.

Sharpening and failure cases.

Sharpening does not consistently improve reference fidelity. At Gaussian $\sigma_n = 20$, the illustrative Laplacian $k = 3$, $\alpha = 0.1$ after the selected Gaussian denoiser changes PSNR from 31.2319 to 31.2305 dB and reduces SSIM from 0.82511 to 0.81441. The selected $k = 7$, $\alpha = 0.1$ chain performs better, showing that aperture and strength must be assessed together.

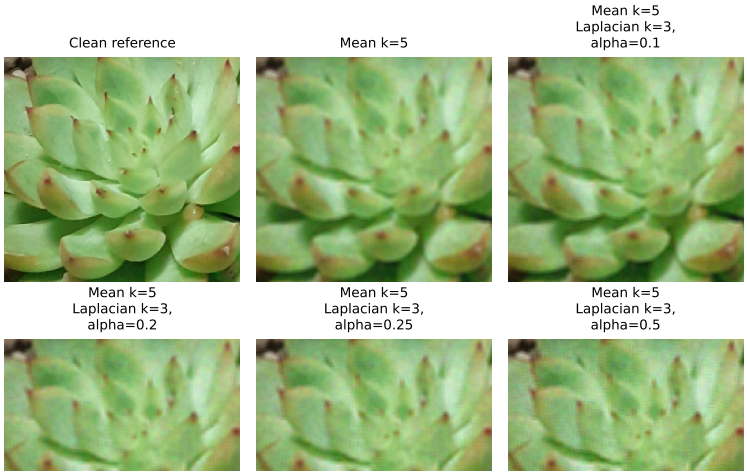


Figure 6: Mean 5×5 followed by Laplacian $k = 3$ with increasing α, at Gaussian $\sigma_n = 20$, seed 42. Stronger sharpening increases residual variation and local edge contrast.

Stronger enhancement makes leaf boundaries harsher and amplifies residual variation. Light/dark overshoot is a halo risk, but these crops do not establish repeated oscillatory ringing. Increased boundary contrast should therefore be compared with the reference rather than automatically interpreted as recovered detail.

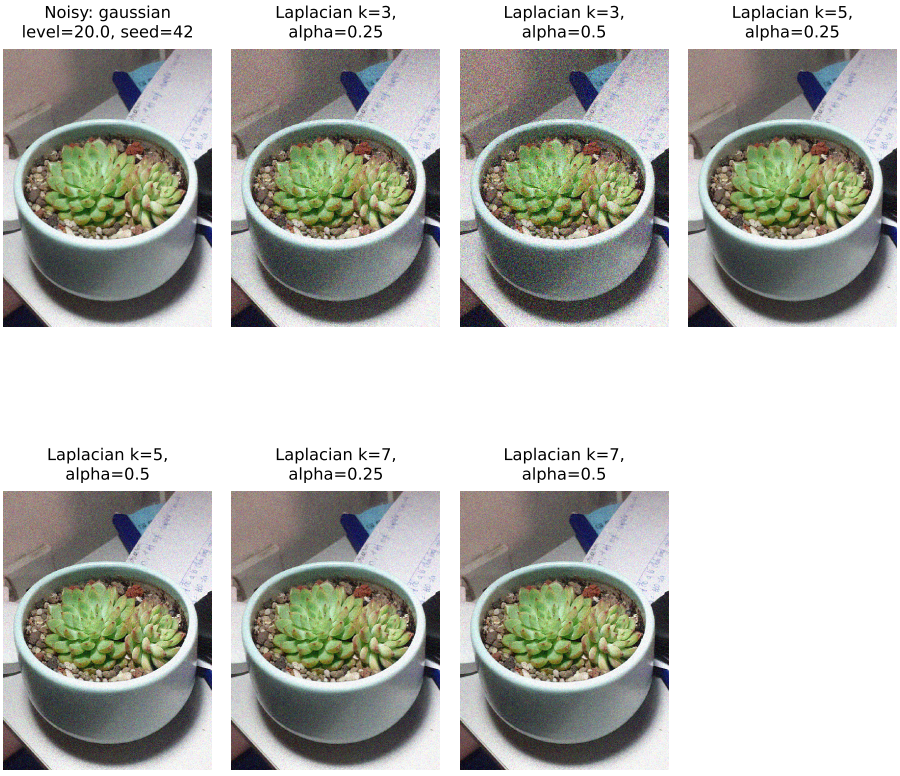


Figure 7: Direct Laplacian sharpening of Gaussian-corrupted input at $\sigma_n = 20$, seed 42. Panels read left to right, top to bottom; labels identify the measured settings.

Applying Laplacian $k = 3$, $\alpha = 0.5$ directly to the noisy image reduces whole-image PSNR from 22.3294 to 16.4111 dB and SSIM from 0.28528 to 0.13111. This failure supports denoising before enhancement. The accompanying results include direct-sharpening

measurements for both noise types, all three apertures, and both tested strengths.

Scope and limitations.

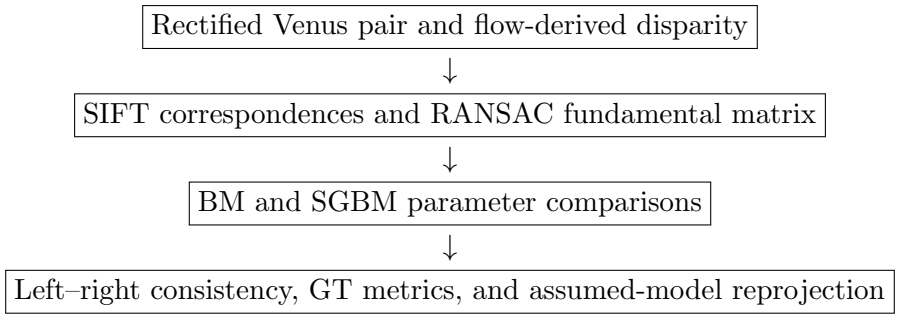
Parameters are selected and evaluated using the same reference image and three seeds. The choices describe the best measured configurations for this image and objective; they do not establish performance on unseen photographs. In the broader search, Gaussian 9×9, $\sigma_f = 1.5$ and 7×7, $\sigma_f = 1.6$ differ by only about 0.0023 dB at $\sigma_n = 30$, so the selected aperture does not imply a substantial perceptual advantage. Resizing and the large smooth background influence the measured trade-offs. Runtime belongs to the recorded environment and includes validation and conversion overhead. Additional images and independent evaluation seeds would help assess generalization.

3 Part B: 3D Reconstruction

3.1 Methodology

3.1.1 Input pair and processing workflow

The experiment uses the Venus sequence from the Middlebury Optical Flow dataset. The left/reference image is `frame10.png`; the right/target image is `frame11.png`. Both are RGB images of size 420×380. They are exact crops of the corresponding rectified Venus stereo views: the left crop starts at $(10, 0)$ and the right crop at $(0, 0)$ in the 434×383 source images. No additional resizing or warping is applied. The accompanying `flow10.flo` field has zero vertical displacement over all 159,600 pixels. For this verified horizontally rectified pair, its horizontal component gives disparity ground truth as $d_{\mathrm{GT}} = -u$, where u is horizontal optical flow. This conversion is specific to the documented pair; an optical-flow file does not generally represent stereo disparity.[4]



3.1.2 Sparse geometry

SIFT detects up to 5,000 features in each image. Brute-force L2 matching retrieves the two nearest neighbours for each query descriptor; a match is retained when $d_1 < 0.75 d_2$. The resulting correspondences estimate a 3×3 fundamental matrix F with USAC_MAGSAC, confidence 0.999, at most 5,000 iterations, and pixel thresholds of 0.5, 1, and 2. The native USAC random state is zero; the project seed is 42. Ground-truth flow is used only for independent diagnostics and disparity evaluation, not to estimate F.[5]

For homogeneous image points $\mathbf{x}_L$ and $\mathbf{x}_R$, corresponding points satisfy

$$\mathbf{x}_R^{\mathsf{T}} F \mathbf{x}_L = 0, \qquad \boldsymbol{\ell}_R = F\mathbf{x}_L, \quad \boldsymbol{\ell}_L = F^{\mathsf{T}}\mathbf{x}_R. \tag{7}$$

The reported symmetric point-to-line error is the mean of the distances in the two images,

$$e_{\mathrm{epi}} = \frac{|\mathbf{x}_R^{\mathsf{T}} F \mathbf{x}_L|}{2} \left(\frac{1}{\sqrt{\ell_{R,1}^2 + \ell_{R,2}^2}} + \frac{1}{\sqrt{\ell_{L,1}^2 + \ell_{L,2}^2}} \right). \tag{8}$$

The Sampson distance is also computed as a first-order geometric approximation. Epipolar lines are drawn in both views for visual inspection. The estimated matrices and inlier masks are retained with the experiment outputs.

[4] Middlebury, *Optical Flow: Other Datasets*, `https://vision.middlebury.edu/flow/data/`. The source stereo views and crop coordinates were checked against the Venus archive at `https://vision.middlebury.edu/stereo/data/scenes2001/data/venus/venus.zip`.

[5] OpenCV, *USAC: Universal RANSAC*, `https://docs.opencv.org/4.x/de/d3e/tutorial_usac.html`; OpenCV, *Camera Calibration and 3D Reconstruction*, `https://docs.opencv.org/4.x/d9/d0c/group__calib3d.html`.

3.1.3 Dense disparity and 3D reprojection

Block Matching (BM) and Semi-Global Block Matching (SGBM) are evaluated with `numDisparities` $\in \{32, 64\}$, `blockSize` $\in \{5, 9, 15\}$, and `minDisparity` $= -16$. BM uses an XSobel prefilter; SGBM uses the three-way mode with $P_1 = 8b^2$ and $P_2 = 32b^2$, where b is the block size. Both use a uniqueness ratio of 10. OpenCV returns fixed-point disparities at 1/16-pixel precision, so each map is divided by 16 before comparison or reprojection. A right-to-left match check retains pixels whose disparity disagreement is at most 1 pixel.[6]

For a calibrated rectified pair, depth is related to disparity by

$$Z = \frac{fB}{d + d_{\text{offs}}}, \qquad X = \frac{(x - c_{x,L})B}{d + d_{\text{offs}}}, \qquad Y = \frac{(y - c_y)B}{d + d_{\text{offs}}}, \tag{9}$$

where f is focal length, B is baseline, and $d_{\text{offs}} = c_{x,R} - c_{x,L}$. No measured calibration was available. The point clouds therefore use an explicitly assumed pinhole model: a 30° horizontal field of view on the source frame, square pixels, zero skew, parallel axes, centered source principal points, and normalized baseline $B = 1$. This gives $f = 434/(2\tan 15°) = 809.855$ pixels, cropped principal points $(206.5, 191)$ and $(216.5, 191)$, and $d_{\text{offs}} = 10$ pixels. The resulting coordinates are in normalized-baseline units and do not establish physical depth accuracy. The field-of-view choice follows Basha et al.; the remaining camera parameters are project assumptions.[7]

3.2 Implementation and Results

3.2.1 Dataset, environment, and evaluation

The input files, source metadata, and result artifacts are recorded in the completed run `20261019T103316Z_9fe8bb55`. The evaluation domain contains 157,906 GT-valid pixels whose flow target remains inside the right image; valid-disparity ratios use all 159,600 image pixels. RMSE and bad-pixel rate are conditional on the intersection of the GT domain and the left–right-consistent prediction mask. A bad pixel has absolute disparity error greater than 1 pixel. No occlusion ground-truth mask is available, so left–right consistency is a reliability estimate rather than an occlusion label.

For prediction d and ground truth d_{GT}, the reported quantities are

$$\text{ValidRatio} = 100\frac{N_{\text{LR}}}{WH}, \quad \text{RMSE} = \sqrt{\frac{1}{N}\sum_{i \in \mathcal{E}}(d_i - d_{\text{GT},i})^2}, \quad \text{BadRate} = 100\frac{N_{\text{bad}}}{N}, \tag{10}$$

where $\mathcal{E}$ is the prediction-valid intersection with the GT evaluation domain, $N = |\mathcal{E}|$, and N_{bad} counts errors greater than 1 pixel. Coverage in Table 7 is $100N/157{,}906$; it is distinct from the all-image valid ratio. Each dense configuration has three warm-ups

[6] OpenCV stereo matching API: `https://docs.opencv.org/5.0/main_modules/stereo.html`. For a review of dense two-frame stereo methods, see D. Scharstein and R. Szeliski, "A Taxonomy and Evaluation of Dense Two-Frame Stereo Correspondence Algorithms," *International Journal of Computer Vision*, vol. 47, pp. 7–42, 2002, `https://vision.middlebury.edu/stereo/taxonomy-IJCV.pdf`.

[7] T. Basha, Y. Moses, and N. Kiryati, "Multi-view Scene Flow Estimation: A View Centered Variational Approach," *International Journal of Computer Vision*, vol. 101, pp. 6–21, 2013, Sec. 4.1, `https://doi.org/10.1007/s11263-012-0542-7`.

and 20 timed repetitions. OpenCV 5.0.0 ran with one thread and OpenCL disabled; the environment used Python 3.13.14 and NumPy 2.5.4. The reported median time includes matcher construction, both matching directions, disparity conversion, and consistency checking; it excludes file I/O, metric computation, figures, and 3D reprojection.

Figure 8: Venus stereo pair and flow-derived disparity. The disparity display is clipped to $[-10, 10]$ pixels for visualization; the original flow values are used for evaluation.

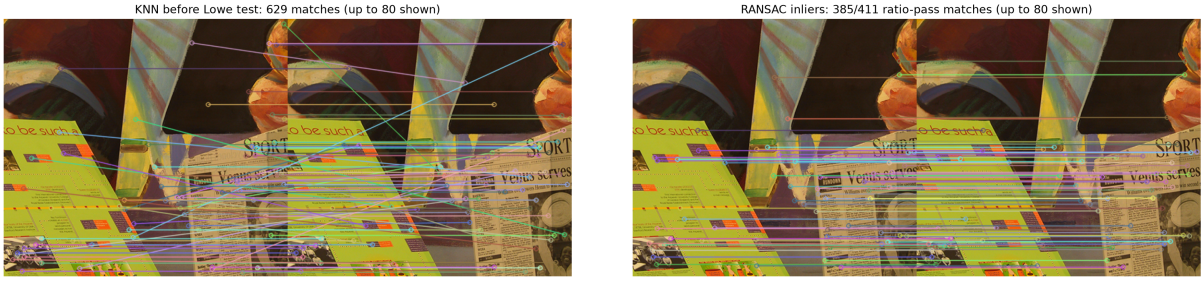
3.2.2 Fundamental matrix and epipolar geometry

The detector returned 629 left and 567 right keypoints. KNN produced 629 candidate pairs; 411 passed the ratio test. Table 6 reports all three RANSAC thresholds. As the threshold increases, more matches are retained as inliers, while the mean error among those inliers also increases. Independent GT-domain checks yield mean symmetric epipolar errors close to 0.11 pixels. The 95th-percentile absolute epipolar-line angle is below 2° in both views for every estimate, a project diagnostic for this rectified pair rather than a grading threshold.

Table 6: Sparse geometry results. Each estimate uses the same 411 ratio-test matches. GT-domain error is evaluated independently on the 157,906 flow-derived correspondences.

Threshold (px)	Inliers	Inlier ratio (%)	Inlier epi. mean (px)	GT-domain epi. mean (px)
0.5	378	91.97	0.0921	0.1150
1.0	385	93.67	0.1049	0.1124
2.0	392	95.38	0.1392	0.1150

Feature detection and matching took 120.743 ms in a single measurement; the three RANSAC fits took 8.828, 3.704, and 3.083 ms at thresholds 0.5, 1, and 2 pixels, respectively. These sparse-stage timings are single samples and are separate from the repeated dense-matching measurements.



(a) Initial descriptor matches.

(b) RANSAC inliers.

Figure 9: SIFT correspondences before and after robust fundamental-matrix estimation.

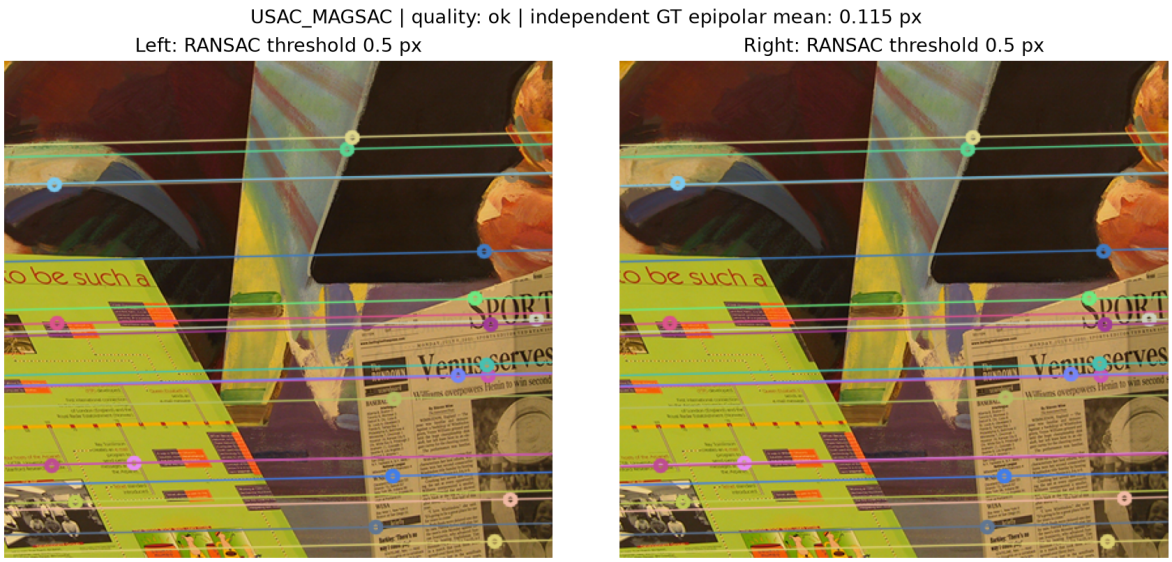


Figure 10: Epipolar lines generated by the 0.5-pixel USAC_MAGSAC estimate, shown in both images. Its independent GT-domain mean symmetric epipolar error is 0.115 pixels.

3.2.3 Dense matching results

Table 7 summarizes all 12 configurations. SGBM provides greater valid coverage and lower conditional error than BM for each tested parameter pair, at a higher measured runtime. Figure 11 and Figure 12 show the left–right-consistent maps. Gray regions are invalid; the color display saturates outside $[-10, 10]$ pixels, while the saved floating-point disparities remain unchanged.

Table 7: Dense disparity comparison. LR-valid ratio is over all image pixels; coverage, RMSE, and bad-pixel rate use the GT-domain intersection with LR-consistent predictions. Runtime is the median of 20 repetitions in milliseconds.

Method / n_{disp} / block	LR-valid ratio (%)	Coverage (%)	RMSE (px)	Bad > 1 px (%)	Median (ms)
BM / 32 / 5	68.483	69.218	1.7335	5.101	15.942
BM / 32 / 9	78.730	79.575	0.8964	2.929	16.663
BM / 32 / 15	82.562	83.448	0.8116	3.322	32.676
BM / 64 / 5	53.749	54.325	3.6564	5.682	36.027
BM / 64 / 9	64.126	64.814	1.0868	3.280	18.381
BM / 64 / 15	68.690	69.427	0.8509	3.733	18.003
SGBM / 32 / 5	90.510	91.481	0.5341	1.521	27.308
SGBM / 32 / 9	90.532	91.503	0.6109	2.011	27.725
SGBM / 32 / 15	90.538	91.510	0.7162	2.851	49.016
SGBM / 64 / 5	76.409	77.228	0.5538	1.697	67.800
SGBM / 64 / 9	76.415	77.235	0.6341	2.266	61.293
SGBM / 64 / 15	76.445	77.266	0.7458	3.236	68.425

3.2.4 Point-cloud reconstruction

For a direct point-cloud comparison, BM and SGBM use `numDisparities` $= 32$ and `blockSize` $= 9$. Reprojection keeps LR-consistent pixels with a positive denominator and finite positive depth. A connected-component filter removes disparity components of at most 100 pixels when adjacent disparity differences are at most 1 pixel; it does not use

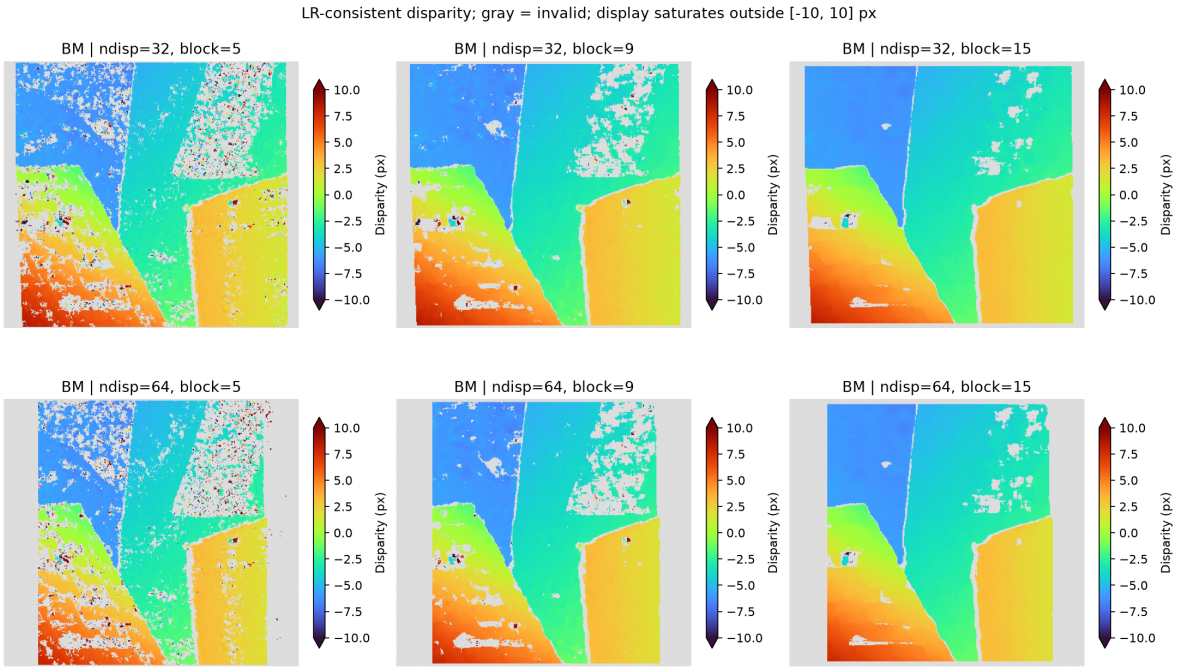


Figure 11: BM left–right-consistent disparity maps for the six tested combinations of disparity range and block size.

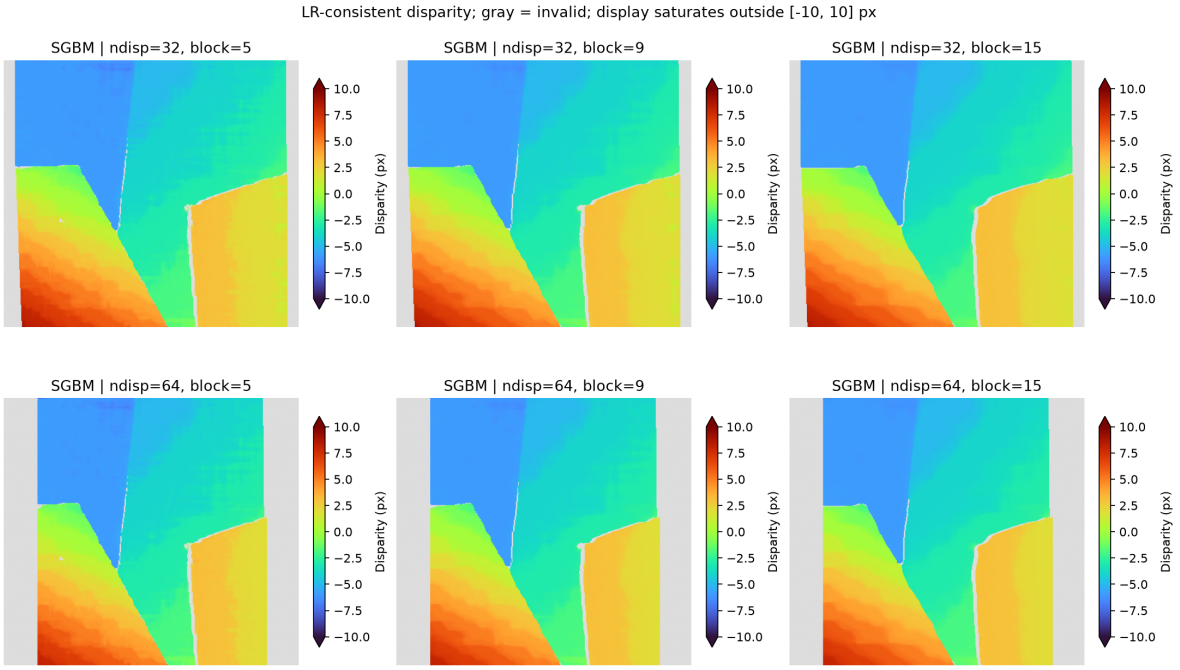


Figure 12: SGBM left–right-consistent disparity maps for the six tested combinations of disparity range and block size.

the GT. The BM cloud contains 125,575 raw points, of which 980 are removed, leaving 124,595. The SGBM cloud contains 144,489 points and the filter removes none. Figure 13 shows the filtered SGBM cloud from three viewpoints; the full point sets are retained in the result artifacts.

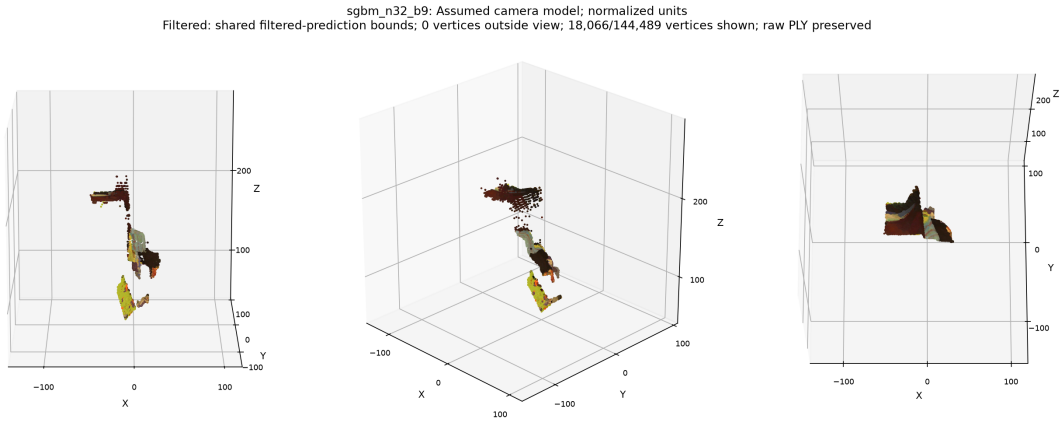


Figure 13: SGBM point cloud for $n_{\text{disp}} = 32$, block size 9, shown from three viewpoints. Axes are in normalized-baseline units under the assumed camera model; the plot samples points for display, while the complete cloud is preserved.

3.3 Comparative Analysis

3.3.1 Accuracy, coverage, and runtime

SGBM's advantage is consistent across the tested grid: its best RMSE is 0.5341 pixels at $n_{\text{disp}} = 32$, block size 5, compared with the best BM RMSE of 0.8116 pixels at $n_{\text{disp}} = 32$, block size 15. SGBM also has the lower bad-pixel rate in each paired setting. At $n_{\text{disp}} = 32$, its LR-valid ratio is about 90.5%, compared with 68.5–82.6% for BM. The difference remains when both methods are evaluated on exactly the same pixels at block size 9: RMSE is 0.5959 pixels for SGBM and 0.8792 pixels for BM; bad-pixel rates are 1.93% and 2.74%, respectively, over the common 125,276-pixel domain. Thus the lower SGBM error is not explained solely by its larger valid region.

This accuracy and coverage come with greater cost in this run. The fastest SGBM case takes 27.308 ms, while the fastest BM case takes 15.942 ms under the same timing protocol. Increasing the disparity range to 64 reduces coverage in both methods for this pair and gives no better RMSE than the best 32-disparity setting. Within SGBM, larger blocks slightly increase coverage but also raise RMSE and bad-pixel rate. BM block size 15 gives the lowest RMSE among BM cases, but its bad-pixel rate is higher than BM at block size 9. These measurements support a trade-off rather than one parameter setting that optimizes every metric.

3.3.2 Visual quality and limitations

The disparity grids show BM's block- and range-dependent invalid areas and speckled responses, especially with a 5-pixel block. Larger BM blocks produce more continuous regions but smooth local variation. SGBM maps are more spatially regular and preserve higher valid coverage, although boundaries are softened and invalid strips remain near some occlusions and image borders. These observations describe the displayed Venus pair; no occlusion ground truth is available to classify every invalid or mismatched pixel.

With assumed focal length, principal points, crop offset, and $B = 1$, these clouds show qualitative normalized depth. Measured camera calibration remains unavailable, so calibrated 3D reconstruction and physical depth accuracy remain unverified. The point-cloud demonstration does not establish completion of the calibrated-reconstruction requirement.

Runtimes apply only to this recorded environment.

4 Part C: Image Stitching

4.1 Methodology

4.1.1 Image sequence and coordinate frame

The experiment uses four ordered, overlapping 1920×1080 PNG images of the interior of Chua Minh Thanh, photographed by the student during a trip to Gia Lai. The input files are `pagoda_01.png` through `pagoda_04.png`; image 3 defines the common coordinate frame. Images 1, 2, and 4 are each mapped directly to image 3, while image 3 uses the identity transform. The inputs are not resized, cropped, or rectified. The run snapshot preserves the source images and their SHA-256 checksums. Figure 14 summarizes the stitching and evaluation stages.

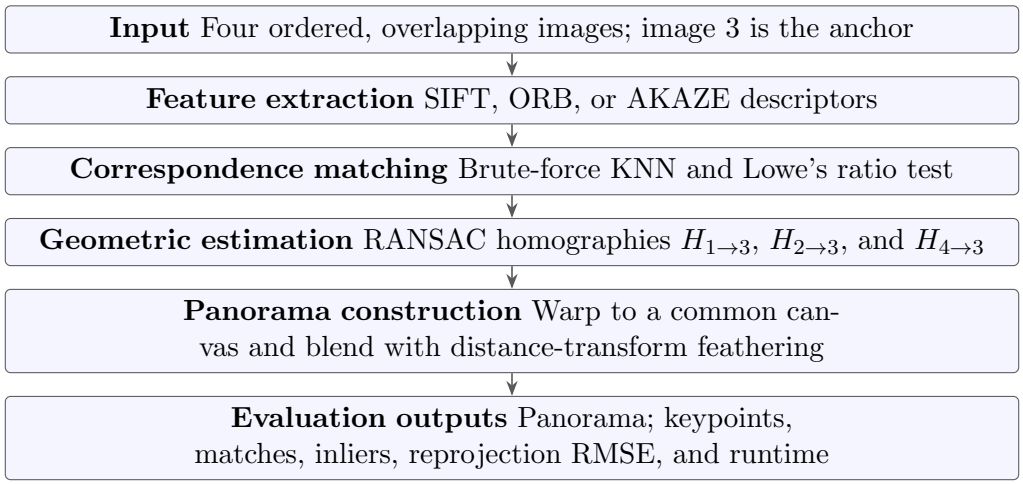


Figure 14: Part C pipeline from the ordered image set to panorama construction and comparative evaluation.

4.1.2 Features, matching, and homography

The experiment compares SIFT, ORB, and AKAZE using OpenCV. SIFT uses up to 5,000 features and the L2 descriptor distance. ORB uses up to 5,000 features, `WTA_K=2`, and Hamming distance. AKAZE uses the oriented MLDB descriptor, threshold 0.001, and a limit of 5,000 features, with Hamming distance. If a detector returns more than the limit, features are stably sorted by response and the corresponding descriptor rows are retained in the same order. Feature extraction runs on grayscale images; geometric warping uses the original colour images.

For each source image and image 3, a brute-force matcher requests the two nearest neighbours, d_1 and d_2. A match passes Lowe's ratio test when

$$d_1 < r d_2, \tag{11}$$

where $r \in \{0.72, 0.75, 0.78\}$. The initial-match count is the number of KNN rows with both neighbours, before applying this test. Descriptor distance is L2 for SIFT and Hamming for ORB and AKAZE.

For each source-to-anchor pair, RANSAC estimates a homography with confidence 0.995,

at most 5,000 iterations, and reprojection threshold $\tau \in \{3, 4, 5, 6\}$ pixels:

$$\lambda \begin{bmatrix} u' \\ v' \\ 1 \end{bmatrix} = H_{i\to 3} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix}, \qquad (u', v') = \left(\frac{q_1}{q_3}, \frac{q_2}{q_3} \right), \tag{12}$$

where $q = H_{i\to 3}[u, v, 1]^T$ and $\lambda = q_3$. The pipeline checks that point sets are non-collinear, the estimated matrix is finite and full rank, at least four inliers remain, and the projected image does not cross a projective horizon. It also checks that the common canvas is finite and contains no more than 40 million pixels. Failed configurations retain their status and reason.

4.1.3 Warping, blending, and evaluation

The main comparison warps all four images into the image-3 frame and uses distance-transform feather weights. Each source colour is multiplied by its weight before warping; the weighted colours and weights are accumulated separately:

$$P(x, y) = \frac{\sum_i \mathcal{W}_i[w_i I_i](x, y)}{\sum_i \mathcal{W}_i[w_i](x, y)}, \tag{13}$$

where $\mathcal{W}_i$ warps a source array into the common canvas and w_i is its source feather weight. Both the premultiplied colour $w_i I_i$ and the weight w_i are warped before accumulation; division is evaluated only on valid support. The support mask comes from the weights, so black image content is not mistaken for an empty region. The output is cropped to the rectangular extent of valid support; unsupported corners within the rectangle remain masked.

Initial matches are counted before the ratio test. The inlier ratio uses all ratio-pass correspondences across the three pairs as its denominator. For pair p, let n_p be the RANSAC inlier count and S_p the sum of squared forward reprojection errors from source to image 3. The pooled measures are

$$\text{InlierRatio} = 100 \frac{\sum_p n_p}{\sum_p M_p}, \qquad \text{RMSE} = \sqrt{\frac{\sum_p S_p}{\sum_p n_p}}, \tag{14}$$

where M_p is the number of ratio-pass matches. RMSE is in pixels; pair RMSE values are not averaged directly. Keypoint totals count each of the four images once. Convex-hull coverage is reported as a spatial-support diagnostic, not as ground-truth alignment accuracy.

Each selected or baseline pipeline is measured after one warm-up and five full repetitions, with extraction and matching repeated each time. Total stitching time runs from detector creation and grayscale conversion through the cropped panorama; it excludes file I/O, metric calculation, and figure generation. The reported value is the median and IQR of five samples. OpenCV runs single-threaded with OpenCL disabled. A separate comparison holds homographies fixed and measures canvas setup, warp, blending, and crop for feathering and multi-band blending with 3, 5, and 7 bands. These stage-only timings are not added to or subtracted from the full-pipeline timings.

4.2 Implementation and Results

4.2.1 Environment and parameter grid

The pipeline uses OpenCV for feature detection, descriptor matching, homography estimation, warping, and blending. Measurements were collected with Python 3.13.14, NumPy 2.5.4, OpenCV 5.0.0 from opencv-contrib-python-headless 5.0.0.93, and Matplotlib 3.11.2 on Linux x86_64. The accompanying reproducibility package preserves the inputs, configurations, source snapshot, homographies, masks, timings, and output checksums. The experiment tested all 12 combinations of three Lowe ratios and four RANSAC thresholds for each detector, giving 36 configurations in total. Every configuration produced a panorama, and all 108 pair records passed validation. The selected settings were reviewed visually within this image sequence and parameter grid; they are not asserted to be universal optima.

Table 8: Detected and retained features per image. Each detector uses at most 5,000 features; SIFT returned 5,001 on image 3 before the stable response limit.

Image	SIFT detected / used	ORB detected / used	AKAZE detected / used
1	5,000 / 5,000	5,000 / 5,000	5,000 / 5,000
2	5,000 / 5,000	5,000 / 5,000	5,000 / 5,000
3	5,001 / 5,000	5,000 / 5,000	5,000 / 5,000
4	5,000 / 5,000	5,000 / 5,000	5,000 / 5,000

(a) SIFT (b) ORB (c) AKAZE

Figure 15: Detected keypoints on image 3. The first 500 retained points are drawn for readability; counts for all four images are in Table 8.

4.2.2 Baseline and selected comparisons

The common baseline is ratio 0.75 and threshold 3 px. Visual review selected SIFT 0.75/5 px, ORB 0.78/5 px, and AKAZE 0.78/5 px. Table 9 gives the pooled counts, forward reprojection error, and repeated total pipeline time. Initial matches precede the ratio test; the ratio-pass count is shown so the inlier-ratio denominator is explicit.

The selected canvas dimensions are 3793×1560 for SIFT, 3779×1554 for ORB, and 3791×1560 for AKAZE. Selected pooled convex-hull support ranges from 55.52% to 69.11% for SIFT, 41.28% to 44.22% for ORB, and 53.26% to 68.55% for AKAZE across the source and anchor views. This diagnostic shows that the accepted points cover different spatial extents; it does not prove absolute alignment accuracy.

Table 9: Baseline and selected end-to-end stitching results. B and S denote baseline and selected settings. Initial matches count complete KNN rows before Lowe's ratio test; P/R are ratio-pass matches and RANSAC inliers. Each configuration has 15,000 initial matches across the three source-to-anchor pairs. Runtime is the median (IQR) in milliseconds over five repetitions after one warm-up.

Detector	Param.	Features D/U	Initial KNN	Matches P/R	Inlier (%)	RMSE (px)	Time (ms)
SIFT	B 0.75/3	20,001/20,000	15,000	6,582/5,927	90.05	0.707	2,898.192 (66.391)
SIFT	S 0.75/5	20,001/20,000	15,000	6,582/5,950	90.40	0.726	2,838.351 (163.371)
ORB	B 0.75/3	20,000/20,000	15,000	5,828/5,332	91.49	1.215	975.079 (39.572)
ORB	S 0.78/5	20,000/20,000	15,000	6,442/6,017	93.40	1.401	1,025.075 (69.450)
AKAZE	B 0.75/3	20,000/20,000	15,000	6,806/6,416	94.27	0.832	2,114.515 (34.142)
AKAZE	S 0.78/5	20,000/20,000	15,000	7,184/6,797	94.61	0.963	1,965.790 (126.001)

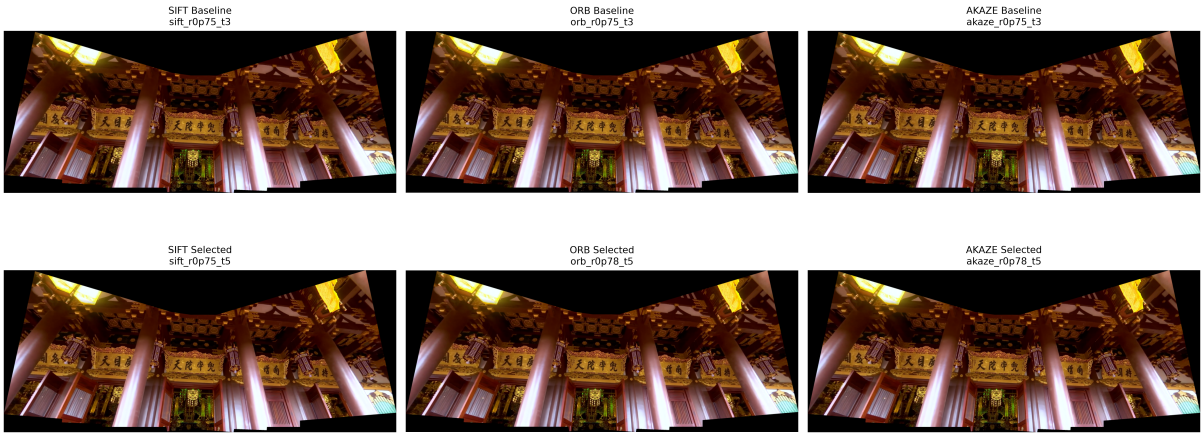


Figure 16: Baseline (top) and selected (bottom) feather-blended panoramas for SIFT, ORB, and AKAZE in the image-3 coordinate frame. Black regions lie outside valid image support.

(a) Initial SIFT KNN matches

(b) SIFT RANSAC inliers

Figure 17: Example initial and inlier correspondences from image 1 to image 3. Up to 80 matches are drawn for readability; full arrays, homographies, and inlier masks are retained in the run.

4.3 Comparative Analysis

4.3.1 Parameter sensitivity and geometric stability

Table 10 compares the tested Lowe-ratio endpoints at a fixed 3 px RANSAC threshold. A larger ratio admits more correspondences and inliers but lowers the accepted inlier percentage. Table 11 compares 3 px and 6 px thresholds at ratio 0.75. Increasing the threshold accepts additional points and increases fitted RMSE. Corner deviation does not change monotonically with fitted RMSE, so the quantities describe different aspects of the fit.

The selected homographies were also refitted from saved correspondences after permutations with NumPy seeds 42, 123, and 2026. Across the three direct source-to-anchor pairs, all 108 homography-and-canvas geometry checks passed. This procedure measures fitted reprojection and projected-corner changes; it does not rerun feature extraction, matching,

Table 10: Lowe-ratio endpoints at fixed RANSAC threshold 3 px. Counts pool the three source-to-anchor pairs.

Detector	Ratio-pass matches	Inliers	Inlier ratio (%)
SIFT	6,300→6,901	5,785→6,078	91.83→88.07
ORB	5,339→6,442	4,930→5,741	92.34→89.12
AKAZE	6,484→7,184	6,168→6,692	95.13→93.15

Table 11: RANSAC-threshold endpoints at fixed Lowe ratio 0.75. Corner change is the maximum projected-corner deviation observed across three correspondence permutations.

Detector	Inliers	RMSE (px)	Max corner change (px)
SIFT	5,927→5,950	0.707→0.723 px	4.038→0.937 px
ORB	5,332→5,546	1.215→1.406 px	20.084→30.301 px
AKAZE	6,416→6,537	0.832→0.983 px	12.224→2.932 px

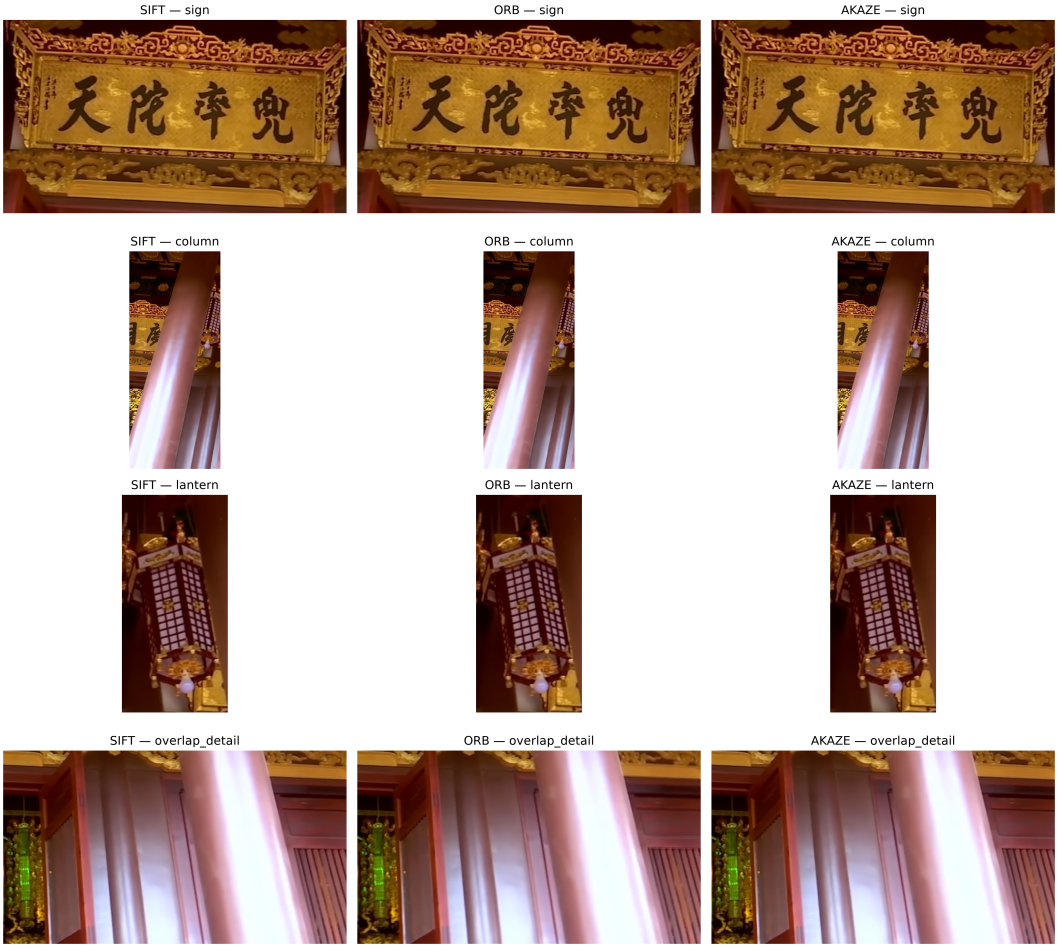


Figure 18: Fixed crops in image-3 coordinates: sign, column, lantern, and overlap detail for the selected detectors.

or blending. Maximum corner changes are 1.565 px for SIFT, 24.328 px for ORB, and 6.038 px for AKAZE. ORB is substantially more sensitive at extrapolated corners despite its high inlier ratio. SIFT gives the lowest selected pooled RMSE and smallest selected corner variation, while AKAZE retains the most selected inliers.

In the inspected native sign crops, lettering and ornament remain continuous across the selected results. Feathering avoids an obvious hard seam at the central sign, while fine overlap detail softens and column ghosting remains visible. The panoramas show leaning outer columns and roof stretching from the planar projection. Bright streaks and saturated yellow highlights persist and are not recovered by changing the blend alone.

4.3.2 Feather and multi-band blending

A supplementary experiment holds the selected homographies, support masks, and crop fixed while comparing feathering with 3-, 5-, and 7-band multi-band blending. Table 12 reports stage-only timing for canvas setup, warping, blending, and cropping. Mean absolute colour change is measured from feathering on common valid support; it is descriptive and not a quality score.

Table 12: Fixed-homography blending comparison. Time is the median (IQR) over five repetitions after one warm-up.

Detector	Blend	Time (ms)	Mean absolute change
SIFT	Feather	815.838 (70.504)	0.000
SIFT	Multi-band, 3	1,001.501 (67.641)	0.662
SIFT	Multi-band, 5	1,028.602 (46.028)	0.779
SIFT	Multi-band, 7	1,027.439 (97.292)	1.315
ORB	Feather	687.850 (208.455)	0.000
ORB	Multi-band, 3	978.913 (58.832)	0.738
ORB	Multi-band, 5	1,114.966 (43.292)	0.854
ORB	Multi-band, 7	1,238.949 (51.768)	1.390
AKAZE	Feather	769.430 (76.210)	0.000
AKAZE	Multi-band, 3	1,196.965 (92.689)	0.659
AKAZE	Multi-band, 5	1,158.733 (41.575)	0.775
AKAZE	Multi-band, 7	1,093.339 (145.709)	1.312

On the inspected sign, the blending options preserve similar lettering. SIFT 5-band blending takes 1,028.602 ms compared with 815.838 ms for feathering in the separate stage-only measurement. Seven bands produce the largest mean colour change for each detector, but neither this change nor the band count shows a quality improvement. Feathering remains the shared baseline for the detector comparison. Multi-band blending does not remove projective stretching or recover saturated detail.

Overall, ORB has the lowest selected total stitching time at 1,025.075 ms, SIFT has the lowest selected pooled reprojection RMSE at 0.726 px and the smallest tested corner variation, and AKAZE retains the most selected inliers at 6,797 (94.61%). These results show distinct speed, fit, and support trade-offs on this four-image sequence. They do not establish performance on other scenes or absolute geometric accuracy.

Figure 19: Sign crops from feathering and multi-band blending at fixed homographies. The blending methods preserve similar central lettering while changing local brightness and texture.

5 Conclusion

Across the tested inputs, classical methods produced clear but task-dependent results. Gaussian filtering gave the strongest PSNR for Gaussian noise, while median filtering best suppressed salt-and-pepper noise; sharpening gains were modest and parameter-sensitive. On the Venus pair, SGBM improved disparity coverage and error over BM at a higher runtime. For stitching, SIFT gave the most stable geometric fit, ORB was fastest but more sensitive at extrapolated corners, and AKAZE retained the most inliers. The reconstructed point clouds remain qualitative because the camera model is assumed.

Further work should use measured stereo calibration and occlusion-aware ground truth, then evaluate additional scenes. Filtering could be tested on more reference images and noise distributions, including edge-aware methods. For stitching, exposure compensation and local or non-planar alignment may address illumination changes and perspective distortion; more sequences are needed to determine whether the observed detector trade-offs generalize.

6 References

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A Appendix: Python Code

A.1 Part A: Image Filtering

The excerpts below show the core Part A algorithms and metrics. They omit input checks and run orchestration; the accompanying source package contains the complete executable implementation.

```
import hashlib
import math
import cv2
import numpy as np
from skimage.metrics import structural_similarity

def make_noisy_image(reference, noise_type, noise_level, seed):
    noise_code = 0 if noise_type == "gaussian" else 1
    level_digest =
       hashlib.sha256(float(noise_level).hex().encode()).digest()
    level_code = int.from_bytes(level_digest[:8], "big")
    rng = np.random.default_rng(np.random.SeedSequence([seed,
       noise_code, level_code]))
    if noise_type == "gaussian":
        noise = rng.normal(0, noise_level, reference.shape)
        return np.rint(np.clip(reference.astype(np.float64) + noise, 0,
           255)).astype(np.uint8)

    height, width = reference.shape[:2]
    count = round(noise_level * height * width)
    indices = rng.choice(height * width, size=count, replace=False)
    salt_count = round(count / 2)
    result = reference.copy()
    flat = result.reshape(-1, 3)
    flat[indices[:salt_count]] = 255
    flat[indices[salt_count:]] = 0
    return result

def denoise(image, method, kernel_size, sigma=None):
```

```
    if method == "mean":
        return cv2.blur(image, (kernel_size, kernel_size),
            borderType=cv2.BORDER_REPLICATE)
    if method == "gaussian":
        return cv2.GaussianBlur(image, (kernel_size, kernel_size),
                                sigmaX=sigma, sigmaY=sigma,
                                borderType=cv2.BORDER_REPLICATE)
    return cv2.medianBlur(image, kernel_size)

def sharpen_laplacian(image, laplacian_kernel_size, alpha):
    working = image.astype(np.float32)
    scale = 2.0 ** (4 - 2 * laplacian_kernel_size)
    laplacian = cv2.Laplacian(working, cv2.CV_32F,
                              ksize=laplacian_kernel_size, scale=scale,
                              borderType=cv2.BORDER_REPLICATE)
    return np.rint(np.clip(working - alpha * laplacian, 0,
        255)).astype(np.uint8)

def compute_metrics(reference, result):
    difference = reference.astype(np.float64) -
        result.astype(np.float64)
    mse = float(np.mean(difference ** 2))
    psnr = math.inf if mse == 0 else 10 * math.log10(255 ** 2 / mse)
    ssim = structural_similarity(
        reference, result, data_range=255, channel_axis=-1,
        gaussian_weights=True, sigma=1.5, use_sample_covariance=False
    )
    return {"mse": mse, "psnr_db": psnr, "ssim": float(ssim)}
```

A.2 Part B: 3D Reconstruction

The excerpt shows feature matching and RANSAC estimation, BM/SGBM disparity computation, ground-truth evaluation, and reprojection. The Venus disparity ground truth is derived from horizontal flow for the verified rectified pair. The camera model is assumed, so the point cloud has normalized baseline units and is qualitative rather than metric.

```
import cv2
import numpy as np

def estimate_fundamental(left, right, threshold):
    sift = cv2.SIFT_create(nfeatures=5000)
    kp_l, des_l = sift.detectAndCompute(left, None)
    kp_r, des_r = sift.detectAndCompute(right, None)
    pairs = cv2.BFMatcher(cv2.NORM_L2).knnMatch(des_l, des_r, k=2)
    good = [m for m, n in pairs if m.distance < 0.75 * n.distance]
    pts_l = np.float32([kp_l[m.queryIdx].pt for m in good])
    pts_r = np.float32([kp_r[m.trainIdx].pt for m in good])
    cv2.setRNGSeed(42)
    F, inliers = cv2.findFundamentalMat(
        pts_l, pts_r, cv2.USAC_MAGSAC, threshold, 0.999, 5000)
    return F, pts_l, pts_r, inliers.ravel().astype(bool)

def matcher(method, disparities, block, minimum):
    if method == "bm":
        stereo = cv2.StereoBM_create(disparities, block)
        stereo.setMinDisparity(minimum)
        stereo.setTextureThreshold(10)
        stereo.setPreFilterType(cv2.STEREO_BM_PREFILTER_XSOBEL)
        stereo.setPreFilterSize(9)
    else:
        stereo = cv2.StereoSGBM_create(
            minimum, disparities, block, P1=8 * block ** 2,
            P2=32 * block ** 2, mode=cv2.STEREO_SGBM_MODE_SGBM_3WAY)
    stereo.setUniquenessRatio(10)
    stereo.setPreFilterCap(31)
    stereo.setSpeckleWindowSize(0)
    stereo.setSpeckleRange(0)
    stereo.setDisp12MaxDiff(1_000_000)
    return stereo

def compute_disparity(left, right, method, disparities, block,
   minimum=-16):
    gray_left = cv2.cvtColor(left, cv2.COLOR_BGR2GRAY)
    gray_right = cv2.cvtColor(right, cv2.COLOR_BGR2GRAY)
    minimum_r = -minimum - disparities + 1
    raw_l = matcher(method, disparities, block,
       minimum).compute(gray_left, gray_right)
    raw_r = matcher(method, disparities, block,
       minimum_r).compute(gray_right, gray_left)
    d_l, d_r = raw_l.astype(np.float32) / 16, raw_r.astype(np.float32)
       / 16
    valid_l = (raw_l > (minimum - 1) * 16) & (d_l >= minimum) & (d_l <
       minimum + disparities)
    valid_r = (raw_r > (minimum_r - 1) * 16) & (d_r >= minimum_r) &
       (d_r < minimum_r + disparities)
    y, x = np.indices(d_l.shape)
    x_r = x - d_l
```

```
    inside = valid_l & (x_r >= 0) & (x_r <= d_l.shape[1] - 1)
    nearest = np.clip(np.rint(x_r).astype(int), 0, d_l.shape[1] - 1)
    consistent = inside & valid_r[y, nearest] & (np.abs(d_l + d_r[y,
       nearest]) <= 1.0)
    return d_l, consistent

def flow_ground_truth(flow):
    valid = np.isfinite(flow).all(axis=2) & (np.abs(flow) <=
       1e9).all(axis=2)
    valid &= np.abs(flow[:, :, 1]) <= 1e-4
    y, x = np.indices(valid.shape)
    x_r, y_r = x + flow[:, :, 0], y + flow[:, :, 1]
    inside = ((x_r >= 0) & (x_r < valid.shape[1]) &
              (y_r >= 0) & (y_r < valid.shape[0]))
    return -flow[:, :, 0].astype(np.float32), valid & inside

def disparity_metrics(disparity, lr_valid, gt_disparity, gt_domain):
    error = np.abs(disparity.astype(np.float64) - gt_disparity)
    values = error[lr_valid & gt_domain]
    rmse = np.sqrt(np.mean(values ** 2)) if values.size else None
    bad_pct = 100 * np.mean(values > 1.0) if values.size else None
    return rmse, bad_pct

def reproject(disparity, valid, reprojection_matrix):
    points = cv2.reprojectImageTo3D(disparity.astype(np.float32),
       reprojection_matrix)
    valid = valid & np.isfinite(points).all(axis=2) & (points[:, :, 2]
       > 0)
    points[~valid] = np.nan
    return points, valid
```

A.3 Part C: Image Stitching

The excerpt shows feature detection, KNN matching with Lowe's ratio test, RANSAC homographies to image 3, and distance-transform feather blending. Input loading, failure handling, and experiment bookkeeping are omitted.

```python
import cv2
import numpy as np

def create_detector(method, maximum=5000):
    if method == "sift":
        return cv2.SIFT_create(nfeatures=maximum), cv2.NORM_L2
    if method == "orb":
        return cv2.ORB_create(nfeatures=maximum, WTA_K=2),
            cv2.NORM_HAMMING
    return (cv2.xfeatures2d.AKAZE_create(threshold=0.001,
        max_points=maximum),
            cv2.NORM_HAMMING)

def extract_features(images, method):
    detector, norm = create_detector(method)
    features = []
    for image in images:
        gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
        keypoints, descriptors = detector.detectAndCompute(gray, None)
        order = sorted(range(len(keypoints)),
                       key=lambda i: -keypoints[i].response)[:5000]
        features.append(([keypoints[i] for i in order],
            descriptors[order]))
    return features, norm

def match_features(source, destination, norm, ratio=0.75):
    key_s, desc_s = source
    key_d, desc_d = destination
    pairs = cv2.BFMatcher(norm, crossCheck=False).knnMatch(desc_s,
        desc_d, k=2)
    complete = [pair for pair in pairs if len(pair) == 2]
    good = [first for first, second in complete
            if first.distance < ratio * second.distance]
    points_s = np.float32([key_s[m.queryIdx].pt for m in good])
    points_d = np.float32([key_d[m.trainIdx].pt for m in good])
    return points_s, points_d, len(complete), len(good)

def estimate_homography(points_s, points_d, threshold):
    H, mask = cv2.findHomography(
        points_s, points_d, cv2.RANSAC, threshold,
        maxIters=5000, confidence=0.995)
    inliers = mask.ravel().astype(bool)
    H = H / H[2, 2]
    projected = cv2.perspectiveTransform(
        points_s[inliers].reshape(-1, 1, 2), H).reshape(-1, 2)
    rmse = np.sqrt(np.mean(np.sum((projected - points_d[inliers]) ** 2,
        axis=1)))
    return H, inliers, rmse

def stitch_four(images, method, ratio=0.75, threshold=3.0):
    features, norm = extract_features(images, method)
    anchor = 2
```

```
    transforms = [np.eye(3) for _ in images]
    pair_results = []
    for index in (0, 1, 3):
        source, destination, initial, passed = match_features(
            features[index], features[anchor], norm, ratio)
        H, inliers, rmse = estimate_homography(source, destination,
            threshold)
        transforms[index] = H
        inlier_count = int(inliers.sum())
        pair_results.append((initial, passed, inlier_count,
                             100 * inlier_count / passed, rmse))
    panorama = feather_blend(images, transforms)
    return panorama, pair_results

def feather_blend(images, transforms):
    corners = []
    for image, H in zip(images, transforms):
        h, w = image.shape[:2]
        xy = np.float32([[0, 0], [w, 0], [w, h], [0, h]]).reshape(-1,
            1, 2)
        corners.append(cv2.perspectiveTransform(xy, H).reshape(-1, 2))
    all_corners = np.concatenate(corners)
    lower = np.floor(all_corners.min(axis=0))
    upper = np.ceil(all_corners.max(axis=0))
    width, height = (upper - lower).astype(int)
    translate = np.array([[1, 0, -lower[0]], [0, 1, -lower[1]], [0, 0,
        1]])
    color_sum = np.zeros((height, width, 3), np.float32)
    weight_sum = np.zeros((height, width), np.float32)
    for image, H in zip(images, transforms):
        mask = np.pad(np.ones(image.shape[:2], np.uint8), 1)
        weight = cv2.distanceTransform(mask, cv2.DIST_L2,
                                       cv2.DIST_MASK_PRECISE)[1:-1,
                                           1:-1]
        weight /= weight.max()
        matrix = translate @ H
        color_sum += cv2.warpPerspective(
            image.astype(np.float32) * weight[:, :, None], matrix,
            (width, height), flags=cv2.INTER_LINEAR)
        weight_sum += cv2.warpPerspective(weight, matrix, (width,
            height))
    valid = weight_sum > 1e-8
    np.divide(
        color_sum, weight_sum[:, :, None], out=color_sum,
        where=valid[:, :, None]
    )
    panorama = np.rint(np.clip(color_sum, 0, 255)).astype(np.uint8)
    y, x = np.nonzero(valid)
    return panorama[y.min():y.max()+1, x.min():x.max()+1]
```

B Part A: Complete Final-Grid Results

These tables reproduce all 528 configuration groups from the final Part A run.

Source: `20261019T024705Z_aef5c7be/summary.csv`.

Quality is mean ± sample standard deviation over seeds 42, 43, and 44. Time is median (IQR), in ms, over 30 samples per processed configuration; noisy inputs have no processing runtime. The earlier parameter-search experiments are outside this final grid. Configuration notation: M = mean, G = Gaussian, Med = median; k is denoising kernel width and s is Gaussian filter sigma. $+L(k, a)$ denotes denoising followed by Laplacian aperture k and strength a; L alone denotes direct sharpening of the noisy input. Laplacian scale is 2^{4-2k}. All kernel sizes are square.

B.1 Gaussian noise: standard deviation 10

Table 13: Complete final-grid results: Gaussian noise, $\sigma_n = 10$.

Configuration	PSNR (dB)	SSIM	MSE	Time (ms)
Noisy input	28.2319 ± 0.0001	0.54723 ± 0.00010	97.70 ± 0.00	N/A
M 3	33.7557 ± 0.0018	0.87724 ± 0.00005	27.38 ± 0.01	0.422 (0.037)
M 3 + L(3, 0.1)	33.6172 ± 0.0012	0.85861 ± 0.00007	28.27 ± 0.01	8.714 (1.424)
M 3 + L(3, 0.2)	33.3516 ± 0.0008	0.83704 ± 0.00007	30.06 ± 0.01	7.967 (2.712)
M 3 + L(3, 0.25)	33.1784 ± 0.0011	0.82555 ± 0.00009	31.28 ± 0.01	7.387 (1.201)
M 3 + L(3, 0.5)	32.0252 ± 0.0029	0.76279 ± 0.00018	40.79 ± 0.03	8.328 (1.495)
M 3 + L(5, 0.1)	33.9140 ± 0.0010	0.87715 ± 0.00002	26.40 ± 0.01	9.435 (1.170)
M 3 + L(5, 0.2)	33.8397 ± 0.0004	0.86593 ± 0.00002	26.86 ± 0.00	10.166 (2.778)
M 3 + L(5, 0.25)	33.8103 ± 0.0004	0.86137 ± 0.00003	27.04 ± 0.00	9.484 (1.177)
M 3 + L(5, 0.5)	33.5633 ± 0.0027	0.84176 ± 0.00006	28.63 ± 0.02	9.707 (1.516)
M 3 + L(7, 0.1)	33.8914 ± 0.0017	0.87799 ± 0.00004	26.54 ± 0.01	10.874 (1.646)
M 3 + L(7, 0.2)	33.9027 ± 0.0005	0.87361 ± 0.00002	26.47 ± 0.00	11.124 (1.182)
M 3 + L(7, 0.25)	33.8703 ± 0.0004	0.86960 ± 0.00002	26.67 ± 0.00	11.668 (1.360)
M 3 + L(7, 0.5)	33.7212 ± 0.0013	0.85517 ± 0.00003	27.60 ± 0.01	10.640 (1.145)
M 5	31.5249 ± 0.0038	0.89101 ± 0.00010	45.77 ± 0.04	0.546 (0.110)
M 7	29.6966 ± 0.0024	0.87008 ± 0.00003	69.73 ± 0.04	1.384 (0.810)
M 9	28.3464 ± 0.0025	0.84862 ± 0.00005	95.16 ± 0.06	1.557 (0.857)
M 11	27.2953 ± 0.0026	0.83092 ± 0.00004	121.21 ± 0.07	1.059 (0.631)
G 3 / 0.6	33.4188 ± 0.0012	0.80630 ± 0.00007	29.59 ± 0.01	0.660 (0.118)
G 3 / 0.8	34.2654 ± 0.0010	0.86351 ± 0.00001	24.35 ± 0.01	0.643 (0.165)
G 3 / 1.0	34.2581 ± 0.0010	0.87654 ± 0.00002	24.39 ± 0.01	0.651 (0.264)
G 3 / 1.2	34.1518 ± 0.0009	0.87962 ± 0.00002	25.00 ± 0.01	0.642 (0.055)
G 3 / 1.4	34.0580 ± 0.0014	0.88008 ± 0.00004	25.54 ± 0.01	0.655 (0.081)
G 3 / 1.5	34.0230 ± 0.0017	0.88004 ± 0.00004	25.75 ± 0.01	0.650 (0.098)
G 3 / 1.6	33.9858 ± 0.0019	0.87988 ± 0.00004	25.97 ± 0.01	0.635 (0.122)
G 5 / 0.6	33.4290 ± 0.0011	0.80741 ± 0.00008	29.52 ± 0.01	1.001 (0.564)
G 5 / 0.8	34.2954 ± 0.0021	0.87793 ± 0.00003	24.18 ± 0.01	0.974 (0.294)
G 5 / 0.8 + L(3, 0.1)	34.2319 ± 0.0006	0.86036 ± 0.00002	24.54 ± 0.00	7.559 (1.023)
G 5 / 0.8 + L(3, 0.2)	34.0034 ± 0.0007	0.83983 ± 0.00003	25.87 ± 0.00	7.880 (1.085)
G 5 / 0.8 + L(3, 0.25)	33.8376 ± 0.0015	0.82901 ± 0.00002	26.87 ± 0.01	8.506 (1.289)
G 5 / 0.8 + L(3, 0.5)	32.6201 ± 0.0036	0.76969 ± 0.00010	35.57 ± 0.03	7.949 (1.465)

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Configuration	PSNR (dB)	SSIM	MSE	Time (ms)
G 5 / 0.8 + L(5, 0.1)	34.4272 ± 0.0010	0.87578 ± 0.00002	23.46 ± 0.01	9.818 (1.629)
G 5 / 0.8 + L(5, 0.2)	34.2656 ± 0.0011	0.86101 ± 0.00002	24.35 ± 0.01	10.877 (2.167)
G 5 / 0.8 + L(5, 0.25)	34.2094 ± 0.0009	0.85572 ± 0.00002	24.67 ± 0.01	9.527 (1.187)
G 5 / 0.8 + L(5, 0.5)	33.7586 ± 0.0031	0.83003 ± 0.00007	27.37 ± 0.02	9.445 (1.136)
G 5 / 0.8 + L(7, 0.1)	34.4378 ± 0.0015	0.87848 ± 0.00002	23.40 ± 0.01	11.814 (1.477)
G 5 / 0.8 + L(7, 0.2)	34.3880 ± 0.0008	0.87169 ± 0.00004	23.67 ± 0.00	11.486 (0.951)
G 5 / 0.8 + L(7, 0.25)	34.3298 ± 0.0008	0.86687 ± 0.00004	23.99 ± 0.00	11.114 (2.406)
G 5 / 0.8 + L(7, 0.5)	34.0744 ± 0.0026	0.84998 ± 0.00008	25.45 ± 0.02	11.765 (2.527)
G 5 / 1.0	33.9775 ± 0.0032	0.90001 ± 0.00006	26.02 ± 0.02	0.993 (0.394)
G 5 / 1.2	33.4499 ± 0.0036	0.90694 ± 0.00009	29.38 ± 0.02	1.004 (0.402)
G 5 / 1.4	33.0227 ± 0.0039	0.90753 ± 0.00010	32.42 ± 0.03	1.096 (0.407)
G 5 / 1.5	32.8198 ± 0.0039	0.90689 ± 0.00010	33.97 ± 0.03	1.054 (0.591)
G 5 / 1.6	32.6960 ± 0.0036	0.90630 ± 0.00009	34.95 ± 0.03	0.956 (0.209)
G 7 / 0.6	33.4290 ± 0.0011	0.80741 ± 0.00008	29.52 ± 0.01	2.185 (0.679)
G 7 / 0.8	34.2954 ± 0.0021	0.87793 ± 0.00003	24.18 ± 0.01	2.330 (0.750)
G 7 / 1.0	33.9042 ± 0.0032	0.90142 ± 0.00006	26.46 ± 0.02	2.348 (0.684)
G 7 / 1.2	33.2199 ± 0.0040	0.90859 ± 0.00010	30.98 ± 0.03	2.131 (0.628)
G 7 / 1.4	32.5513 ± 0.0038	0.90842 ± 0.00009	36.14 ± 0.03	2.206 (0.683)
G 7 / 1.5	32.2570 ± 0.0037	0.90713 ± 0.00009	38.67 ± 0.03	2.165 (0.384)
G 7 / 1.6	32.0077 ± 0.0036	0.90543 ± 0.00009	40.96 ± 0.03	2.301 (0.701)
G 9 / 0.6	33.4290 ± 0.0011	0.80741 ± 0.00008	29.52 ± 0.01	2.678 (0.757)
G 9 / 0.8	34.2954 ± 0.0021	0.87793 ± 0.00003	24.18 ± 0.01	2.975 (0.683)
G 9 / 1.0	33.9042 ± 0.0032	0.90142 ± 0.00006	26.46 ± 0.02	2.727 (0.478)
G 9 / 1.2	33.2199 ± 0.0040	0.90859 ± 0.00010	30.98 ± 0.03	2.551 (1.055)
G 9 / 1.4	32.4436 ± 0.0038	0.90836 ± 0.00010	37.04 ± 0.03	2.907 (1.018)
G 9 / 1.5	32.0982 ± 0.0039	0.90644 ± 0.00009	40.11 ± 0.04	2.813 (0.687)
G 9 / 1.6	31.7629 ± 0.0037	0.90415 ± 0.00009	43.33 ± 0.04	2.774 (0.649)
G 11 / 0.6	33.4290 ± 0.0011	0.80741 ± 0.00008	29.52 ± 0.01	3.222 (0.580)
G 11 / 0.8	34.2954 ± 0.0021	0.87793 ± 0.00003	24.18 ± 0.01	2.900 (0.800)
G 11 / 1.0	33.9042 ± 0.0032	0.90142 ± 0.00006	26.46 ± 0.02	3.145 (0.842)
G 11 / 1.2	33.2199 ± 0.0040	0.90859 ± 0.00010	30.98 ± 0.03	3.191 (0.522)
G 11 / 1.4	32.4090 ± 0.0040	0.90800 ± 0.00010	37.34 ± 0.03	3.197 (1.170)
G 11 / 1.5	32.0982 ± 0.0039	0.90644 ± 0.00009	40.11 ± 0.04	3.103 (0.736)
G 11 / 1.6	31.7198 ± 0.0039	0.90380 ± 0.00010	43.76 ± 0.04	3.293 (0.723)
Med 3	33.5676 ± 0.0021	0.84035 ± 0.00002	28.60 ± 0.01	0.325 (0.050)
Med 3 + L(3, 0.1)	32.9876 ± 0.0015	0.81017 ± 0.00009	32.68 ± 0.01	7.516 (1.068)
Med 3 + L(3, 0.2)	32.2549 ± 0.0011	0.77700 ± 0.00017	38.69 ± 0.01	7.951 (1.624)
Med 3 + L(3, 0.25)	31.8583 ± 0.0010	0.75991 ± 0.00019	42.39 ± 0.01	7.749 (1.630)
Med 3 + L(3, 0.5)	29.8252 ± 0.0025	0.67309 ± 0.00029	67.70 ± 0.04	8.601 (2.033)
Med 3 + L(5, 0.1)	33.5483 ± 0.0024	0.83679 ± 0.00003	28.72 ± 0.02	10.355 (2.087)
Med 3 + L(5, 0.2)	33.2783 ± 0.0025	0.82211 ± 0.00002	30.57 ± 0.02	10.355 (1.770)
Med 3 + L(5, 0.25)	33.1613 ± 0.0026	0.81651 ± 0.00004	31.40 ± 0.02	10.491 (1.668)
Med 3 + L(5, 0.5)	32.4543 ± 0.0038	0.78937 ± 0.00007	36.95 ± 0.03	9.522 (0.969)
Med 3 + L(7, 0.1)	33.6192 ± 0.0027	0.84047 ± 0.00002	28.26 ± 0.02	10.779 (1.082)
Med 3 + L(7, 0.2)	33.4700 ± 0.0029	0.83259 ± 0.00003	29.25 ± 0.02	11.188 (1.242)
Med 3 + L(7, 0.25)	33.3788 ± 0.0024	0.82785 ± 0.00002	29.87 ± 0.02	10.575 (0.912)
Med 3 + L(7, 0.5)	32.9519 ± 0.0037	0.81049 ± 0.00006	32.95 ± 0.03	11.212 (1.282)
Med 5	32.9465 ± 0.0052	0.88819 ± 0.00012	32.99 ± 0.04	3.193 (0.597)
Med 7	31.5483 ± 0.0047	0.88520 ± 0.00001	45.53 ± 0.05	57.561 (0.808)

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Configuration	PSNR (dB)	SSIM	MSE	Time (ms)
Med 9	30.2940 ± 0.0031	0.87387 ± 0.00003	60.77 ± 0.04	61.314 (0.921)
Med 11	29.2400 ± 0.0031	0.86214 ± 0.00001	77.46 ± 0.06	65.872 (0.798)
L(3, 0.25)	24.5459 ± 0.0011	0.38532 ± 0.00018	228.29 ± 0.06	7.700 (1.608)
L(3, 0.5)	21.8661 ± 0.0015	0.28657 ± 0.00021	423.13 ± 0.14	7.057 (1.703)
L(5, 0.25)	27.3845 ± 0.0008	0.51214 ± 0.00007	118.75 ± 0.02	9.627 (1.216)
L(5, 0.5)	26.4934 ± 0.0016	0.47762 ± 0.00005	145.79 ± 0.05	9.182 (1.029)
L(7, 0.25)	27.8633 ± 0.0008	0.53345 ± 0.00008	106.35 ± 0.02	10.493 (1.265)
L(7, 0.5)	27.4368 ± 0.0014	0.51874 ± 0.00005	117.33 ± 0.04	10.580 (1.345)

B.2 Gaussian noise: standard deviation 20

Table 14: Complete final-grid results: Gaussian noise, $\sigma_n = 20$.

Configuration	PSNR (dB)	SSIM	MSE	Time (ms)
Noisy input	22.3294 ± 0.0008	0.28528 ± 0.00013	380.31 ± 0.07	N/A
M 3	30.3019 ± 0.0077	0.72094 ± 0.00009	60.66 ± 0.11	0.427 (0.055)
M 5	30.4018 ± 0.0064	0.83307 ± 0.00013	59.28 ± 0.09	0.532 (0.041)
M 5 + L(3, 0.1)	30.3114 ± 0.0061	0.81635 ± 0.00012	60.53 ± 0.08	7.750 (1.089)
M 5 + L(3, 0.2)	30.1803 ± 0.0062	0.79673 ± 0.00012	62.38 ± 0.09	7.674 (1.332)
M 5 + L(3, 0.25)	30.1035 ± 0.0058	0.78645 ± 0.00010	63.49 ± 0.08	7.503 (0.959)
M 5 + L(3, 0.5)	29.6050 ± 0.0047	0.73014 ± 0.00007	71.22 ± 0.08	7.684 (1.322)
M 5 + L(5, 0.1)	30.4642 ± 0.0063	0.83321 ± 0.00013	58.43 ± 0.09	9.642 (1.140)
M 5 + L(5, 0.2)	30.4426 ± 0.0064	0.82421 ± 0.00015	58.72 ± 0.09	10.083 (1.354)
M 5 + L(5, 0.25)	30.4289 ± 0.0066	0.81963 ± 0.00016	58.91 ± 0.09	9.822 (1.246)
M 5 + L(5, 0.5)	30.3811 ± 0.0063	0.80282 ± 0.00017	59.56 ± 0.09	10.116 (1.901)
M 5 + L(7, 0.1)	30.4652 ± 0.0064	0.83365 ± 0.00013	58.42 ± 0.09	11.465 (3.346)
M 5 + L(7, 0.2)	30.5114 ± 0.0067	0.83179 ± 0.00015	57.80 ± 0.09	10.909 (0.996)
M 5 + L(7, 0.25)	30.5091 ± 0.0066	0.82876 ± 0.00015	57.83 ± 0.09	11.161 (1.397)
M 5 + L(7, 0.5)	30.5115 ± 0.0068	0.81640 ± 0.00017	57.80 ± 0.09	11.056 (0.979)
M 7	29.2136 ± 0.0048	0.84544 ± 0.00011	77.93 ± 0.09	1.861 (0.352)
M 9	28.0738 ± 0.0042	0.83630 ± 0.00009	101.32 ± 0.10	1.134 (0.840)
M 11	27.1102 ± 0.0030	0.82397 ± 0.00008	126.49 ± 0.09	1.868 (0.609)
G 3 / 0.6	28.0822 ± 0.0063	0.56190 ± 0.00011	101.13 ± 0.15	0.714 (0.103)
G 3 / 0.8	29.7683 ± 0.0082	0.67193 ± 0.00015	68.59 ± 0.13	0.708 (0.219)
G 3 / 1.0	30.2195 ± 0.0085	0.70634 ± 0.00014	61.82 ± 0.12	0.699 (0.360)
G 3 / 1.2	30.3404 ± 0.0083	0.71743 ± 0.00011	60.12 ± 0.12	0.693 (0.096)
G 3 / 1.4	30.3679 ± 0.0084	0.72111 ± 0.00012	59.74 ± 0.12	0.681 (0.161)
G 3 / 1.5	30.3702 ± 0.0083	0.72187 ± 0.00011	59.71 ± 0.11	0.637 (0.152)
G 3 / 1.6	30.3690 ± 0.0082	0.72238 ± 0.00011	59.73 ± 0.11	0.659 (0.173)
G 5 / 0.6	28.1156 ± 0.0062	0.56364 ± 0.00010	100.35 ± 0.14	1.311 (0.711)
G 5 / 0.8	30.2461 ± 0.0086	0.70649 ± 0.00014	61.44 ± 0.12	1.164 (0.436)
G 5 / 1.0	31.0099 ± 0.0095	0.77667 ± 0.00014	51.53 ± 0.11	1.153 (0.694)
G 5 / 1.2	31.2134 ± 0.0091	0.81391 ± 0.00012	49.17 ± 0.10	1.208 (0.402)
G 5 / 1.4	31.1810 ± 0.0084	0.83024 ± 0.00013	49.54 ± 0.10	1.267 (0.461)
G 5 / 1.5	31.1289 ± 0.0084	0.83537 ± 0.00013	50.14 ± 0.10	1.330 (0.731)
G 5 / 1.6	31.0883 ± 0.0084	0.83786 ± 0.00013	50.61 ± 0.10	1.449 (0.567)
G 7 / 0.6	28.1156 ± 0.0062	0.56364 ± 0.00010	100.35 ± 0.14	2.495 (0.408)
G 7 / 0.8	30.2461 ± 0.0086	0.70649 ± 0.00014	61.44 ± 0.12	2.532 (0.501)

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Configuration	PSNR (dB)	SSIM	MSE	Time (ms)
G 7 / 1.0	31.0562 ± 0.0092	0.78261 ± 0.00013	50.99 ± 0.11	2.543 (0.508)
G 7 / 1.2	31.2319 ± 0.0089	0.82511 ± 0.00014	48.97 ± 0.10	2.456 (0.419)
G 7 / 1.2 + L(3, 0.1)	31.2305 ± 0.0090	0.81441 ± 0.00016	48.98 ± 0.10	8.415 (0.782)
G 7 / 1.2 + L(3, 0.2)	31.1507 ± 0.0090	0.79843 ± 0.00014	49.89 ± 0.10	8.506 (1.545)
G 7 / 1.2 + L(3, 0.25)	31.1070 ± 0.0089	0.79127 ± 0.00014	50.39 ± 0.10	8.380 (0.920)
G 7 / 1.2 + L(3, 0.5)	30.7506 ± 0.0084	0.75290 ± 0.00014	54.70 ± 0.11	8.693 (1.168)
G 7 / 1.2 + L(5, 0.1)	31.2959 ± 0.0088	0.82271 ± 0.00013	48.25 ± 0.10	10.546 (0.802)
G 7 / 1.2 + L(5, 0.2)	31.2026 ± 0.0091	0.80808 ± 0.00015	49.30 ± 0.10	10.740 (1.262)
G 7 / 1.2 + L(5, 0.25)	31.1779 ± 0.0088	0.80297 ± 0.00014	49.58 ± 0.10	10.396 (0.563)
G 7 / 1.2 + L(5, 0.5)	30.9967 ± 0.0088	0.77881 ± 0.00013	51.69 ± 0.10	10.536 (0.577)
G 7 / 1.2 + L(7, 0.1)	31.3128 ± 0.0087	0.82536 ± 0.00014	48.06 ± 0.10	12.552 (1.664)
G 7 / 1.2 + L(7, 0.2)	31.2478 ± 0.0093	0.81530 ± 0.00016	48.79 ± 0.10	12.621 (1.790)
G 7 / 1.2 + L(7, 0.25)	31.2168 ± 0.0090	0.81023 ± 0.00015	49.14 ± 0.10	12.483 (0.997)
G 7 / 1.2 + L(7, 0.5)	31.1021 ± 0.0091	0.79219 ± 0.00016	50.45 ± 0.11	12.183 (1.026)
G 7 / 1.4	31.1054 ± 0.0085	0.84804 ± 0.00015	50.41 ± 0.10	2.533 (0.473)
G 7 / 1.5	30.9952 ± 0.0081	0.85477 ± 0.00014	51.71 ± 0.10	2.523 (0.463)
G 7 / 1.6	30.8810 ± 0.0079	0.85902 ± 0.00013	53.09 ± 0.10	2.382 (0.562)
G 9 / 0.6	28.1156 ± 0.0062	0.56364 ± 0.00010	100.35 ± 0.14	2.925 (0.734)
G 9 / 0.8	30.2461 ± 0.0086	0.70649 ± 0.00014	61.44 ± 0.12	2.851 (0.518)
G 9 / 1.0	31.0562 ± 0.0092	0.78261 ± 0.00013	50.99 ± 0.11	3.005 (0.741)
G 9 / 1.2	31.2319 ± 0.0089	0.82511 ± 0.00014	48.97 ± 0.10	2.965 (0.597)
G 9 / 1.4	31.0698 ± 0.0084	0.85068 ± 0.00015	50.83 ± 0.10	2.800 (0.702)
G 9 / 1.5	30.9247 ± 0.0079	0.85749 ± 0.00013	52.55 ± 0.10	2.851 (0.656)
G 9 / 1.6	30.7575 ± 0.0074	0.86272 ± 0.00013	54.62 ± 0.09	2.917 (0.506)
G 11 / 0.6	28.1156 ± 0.0062	0.56364 ± 0.00010	100.35 ± 0.14	3.559 (0.939)
G 11 / 0.8	30.2461 ± 0.0086	0.70649 ± 0.00014	61.44 ± 0.12	3.416 (1.164)
G 11 / 1.0	31.0562 ± 0.0092	0.78261 ± 0.00013	50.99 ± 0.11	3.375 (0.715)
G 11 / 1.2	31.2319 ± 0.0089	0.82511 ± 0.00014	48.97 ± 0.10	3.620 (0.983)
G 11 / 1.4	31.0542 ± 0.0083	0.85114 ± 0.00013	51.01 ± 0.10	3.475 (0.791)
G 11 / 1.5	30.9247 ± 0.0079	0.85749 ± 0.00013	52.55 ± 0.10	3.486 (1.112)
G 11 / 1.6	30.7340 ± 0.0073	0.86334 ± 0.00012	54.91 ± 0.09	3.448 (0.984)
Med 3	29.1024 ± 0.0094	0.63952 ± 0.00008	79.95 ± 0.17	0.335 (0.029)
Med 5	30.6926 ± 0.0124	0.79309 ± 0.00037	55.44 ± 0.16	3.226 (0.542)
Med 5 + L(3, 0.1)	30.4096 ± 0.0126	0.76573 ± 0.00039	59.17 ± 0.17	10.087 (1.648)
Med 5 + L(3, 0.2)	30.0499 ± 0.0127	0.73568 ± 0.00042	64.28 ± 0.19	9.734 (1.411)
Med 5 + L(3, 0.25)	29.8493 ± 0.0128	0.72030 ± 0.00043	67.32 ± 0.20	9.611 (1.298)
Med 5 + L(3, 0.5)	28.7124 ± 0.0124	0.64222 ± 0.00050	87.47 ± 0.25	10.231 (2.127)
Med 5 + L(5, 0.1)	30.7054 ± 0.0128	0.79019 ± 0.00038	55.28 ± 0.16	11.762 (0.858)
Med 5 + L(5, 0.2)	30.5818 ± 0.0129	0.77664 ± 0.00038	56.87 ± 0.17	11.972 (1.297)
Med 5 + L(5, 0.25)	30.5326 ± 0.0131	0.77132 ± 0.00039	57.52 ± 0.17	12.056 (1.183)
Med 5 + L(5, 0.5)	30.2547 ± 0.0134	0.74724 ± 0.00042	61.32 ± 0.19	12.059 (1.110)
Med 5 + L(7, 0.1)	30.7437 ± 0.0124	0.79337 ± 0.00037	54.79 ± 0.16	13.442 (0.807)
Med 5 + L(7, 0.2)	30.6945 ± 0.0132	0.78697 ± 0.00038	55.42 ± 0.17	13.266 (0.777)
Med 5 + L(7, 0.25)	30.6574 ± 0.0132	0.78276 ± 0.00038	55.89 ± 0.17	13.476 (1.005)
Med 5 + L(7, 0.5)	30.5084 ± 0.0138	0.76736 ± 0.00041	57.84 ± 0.18	13.843 (1.767)
Med 7	30.3050 ± 0.0095	0.83535 ± 0.00015	60.62 ± 0.13	60.356 (1.466)
Med 9	29.4665 ± 0.0064	0.84220 ± 0.00011	73.52 ± 0.11	63.722 (1.554)
Med 11	28.6174 ± 0.0036	0.83884 ± 0.00009	89.40 ± 0.07	68.367 (1.108)
L(3, 0.25)	18.8547 ± 0.0018	0.18284 ± 0.00012	846.47 ± 0.35	6.910 (1.096)

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Configuration	PSNR (dB)	SSIM	MSE	Time (ms)
L(3, 0.5)	16.4111 ± 0.0024	0.13111 ± 0.00009	1485.84 ± 0.81	6.816 (0.644)
L(5, 0.25)	21.5522 ± 0.0008	0.26207 ± 0.00009	454.84 ± 0.09	9.239 (1.714)
L(5, 0.5)	20.7894 ± 0.0011	0.24084 ± 0.00006	542.17 ± 0.14	9.205 (1.483)
L(7, 0.25)	21.9990 ± 0.0006	0.27661 ± 0.00011	410.37 ± 0.05	10.423 (0.769)
L(7, 0.5)	21.6527 ± 0.0007	0.26768 ± 0.00009	444.43 ± 0.07	10.427 (1.347)

B.3 Gaussian noise: standard deviation 30

Table 15: Complete final-grid results: Gaussian noise, $\sigma_n = 30$.

Configuration	PSNR (dB)	SSIM	MSE	Time (ms)
Noisy input	18.9570 ± 0.0035	0.17793 ± 0.00011	826.76 ± 0.67	N/A
M 3	27.5194 ± 0.0026	0.57881 ± 0.00031	115.12 ± 0.07	0.412 (0.022)
M 5	28.9585 ± 0.0025	0.75971 ± 0.00012	82.65 ± 0.05	0.540 (0.032)
M 5 + L(3, 0.1)	28.7515 ± 0.0024	0.73091 ± 0.00013	86.68 ± 0.05	7.535 (1.187)
M 5 + L(3, 0.2)	28.5059 ± 0.0024	0.70018 ± 0.00015	91.73 ± 0.05	7.626 (1.320)
M 5 + L(3, 0.25)	28.3694 ± 0.0017	0.68432 ± 0.00017	94.65 ± 0.04	7.610 (1.533)
M 5 + L(3, 0.5)	27.5871 ± 0.0014	0.60446 ± 0.00018	113.34 ± 0.04	7.600 (1.016)
M 5 + L(5, 0.1)	28.9540 ± 0.0026	0.75562 ± 0.00013	82.73 ± 0.05	9.532 (1.073)
M 5 + L(5, 0.2)	28.8503 ± 0.0028	0.73984 ± 0.00014	84.73 ± 0.05	9.748 (1.349)
M 5 + L(5, 0.25)	28.8151 ± 0.0027	0.73427 ± 0.00015	85.42 ± 0.05	9.670 (0.915)
M 5 + L(5, 0.5)	28.6158 ± 0.0031	0.70770 ± 0.00016	89.43 ± 0.06	9.622 (0.800)
M 5 + L(7, 0.1)	28.9930 ± 0.0026	0.75970 ± 0.00012	81.99 ± 0.05	11.437 (1.443)
M 5 + L(7, 0.2)	28.9379 ± 0.0029	0.75063 ± 0.00012	83.04 ± 0.06	11.144 (1.738)
M 5 + L(7, 0.25)	28.9090 ± 0.0030	0.74576 ± 0.00013	83.60 ± 0.06	11.222 (0.942)
M 5 + L(7, 0.5)	28.8088 ± 0.0033	0.72884 ± 0.00015	85.55 ± 0.07	11.386 (0.995)
M 7	28.4427 ± 0.0009	0.81002 ± 0.00010	93.07 ± 0.02	1.674 (0.762)
M 9	27.5874 ± 0.0018	0.81749 ± 0.00008	113.33 ± 0.05	1.722 (0.768)
M 11	26.7587 ± 0.0018	0.81285 ± 0.00015	137.15 ± 0.06	1.874 (0.439)
G 3 / 0.6	24.7921 ± 0.0006	0.40221 ± 0.00014	215.71 ± 0.03	0.679 (0.209)
G 3 / 0.8	26.6606 ± 0.0016	0.51725 ± 0.00025	140.29 ± 0.05	0.661 (0.148)
G 3 / 1.0	27.2382 ± 0.0021	0.55866 ± 0.00029	122.82 ± 0.06	0.679 (0.142)
G 3 / 1.2	27.4301 ± 0.0024	0.57289 ± 0.00029	117.51 ± 0.07	0.654 (0.197)
G 3 / 1.4	27.4990 ± 0.0026	0.57795 ± 0.00030	115.66 ± 0.07	0.651 (0.269)
G 3 / 1.5	27.5140 ± 0.0025	0.57902 ± 0.00030	115.26 ± 0.07	0.689 (0.171)
G 3 / 1.6	27.5253 ± 0.0027	0.57980 ± 0.00030	114.96 ± 0.07	0.694 (0.206)
G 5 / 0.6	24.8300 ± 0.0008	0.40379 ± 0.00014	213.84 ± 0.04	1.338 (0.554)
G 5 / 0.8	27.2607 ± 0.0022	0.55770 ± 0.00027	122.18 ± 0.06	1.480 (0.747)
G 5 / 1.0	28.4087 ± 0.0035	0.65156 ± 0.00030	93.80 ± 0.07	1.360 (0.609)
G 5 / 1.2	28.9692 ± 0.0038	0.71013 ± 0.00028	82.44 ± 0.07	1.423 (0.767)
G 5 / 1.4	29.1765 ± 0.0039	0.73957 ± 0.00025	78.60 ± 0.07	1.299 (0.644)
G 5 / 1.5	29.2268 ± 0.0042	0.74984 ± 0.00023	77.70 ± 0.07	1.315 (0.687)
G 5 / 1.6	29.2456 ± 0.0038	0.75513 ± 0.00021	77.36 ± 0.07	1.102 (0.702)
G 7 / 0.6	24.8300 ± 0.0008	0.40379 ± 0.00014	213.84 ± 0.04	2.378 (0.398)
G 7 / 0.8	27.2607 ± 0.0022	0.55770 ± 0.00027	122.18 ± 0.06	2.563 (0.596)
G 7 / 1.0	28.5066 ± 0.0035	0.66028 ± 0.00029	91.71 ± 0.07	2.380 (0.381)
G 7 / 1.2	29.1329 ± 0.0042	0.72885 ± 0.00025	79.40 ± 0.08	2.441 (0.312)
G 7 / 1.4	29.3800 ± 0.0036	0.77297 ± 0.00021	75.00 ± 0.06	2.474 (0.360)

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Configuration	PSNR (dB)	SSIM	MSE	Time (ms)
G 7 / 1.5	29.4183 ± 0.0035	0.78791 ± 0.00016	74.35 ± 0.06	2.526 (0.421)
G 7 / 1.6	29.4204 ± 0.0031	0.79849 ± 0.00016	74.31 ± 0.05	2.380 (0.482)
G 9 / 0.6	24.8300 ± 0.0008	0.40379 ± 0.00014	213.84 ± 0.04	2.974 (0.918)
G 9 / 0.8	27.2607 ± 0.0022	0.55770 ± 0.00027	122.18 ± 0.06	3.062 (0.713)
G 9 / 1.0	28.5066 ± 0.0035	0.66028 ± 0.00029	91.71 ± 0.07	3.133 (0.873)
G 9 / 1.2	29.1329 ± 0.0042	0.72885 ± 0.00025	79.40 ± 0.08	3.045 (1.310)
G 9 / 1.4	29.4005 ± 0.0035	0.77824 ± 0.00020	74.65 ± 0.06	3.027 (0.918)
G 9 / 1.5	29.4227 ± 0.0033	0.79417 ± 0.00018	74.27 ± 0.06	3.045 (0.677)
G 9 / 1.5 + L(3, 0.1)	29.4366 ± 0.0035	0.78818 ± 0.00019	74.03 ± 0.06	8.844 (0.922)
G 9 / 1.5 + L(3, 0.2)	29.3739 ± 0.0037	0.77374 ± 0.00022	75.11 ± 0.06	8.858 (1.131)
G 9 / 1.5 + L(3, 0.25)	29.3521 ± 0.0040	0.76829 ± 0.00024	75.49 ± 0.07	8.789 (1.191)
G 9 / 1.5 + L(3, 0.5)	29.1842 ± 0.0045	0.73996 ± 0.00026	78.46 ± 0.08	8.856 (1.107)
G 9 / 1.5 + L(5, 0.1)	29.4675 ± 0.0035	0.79285 ± 0.00017	73.51 ± 0.06	10.983 (1.170)
G 9 / 1.5 + L(5, 0.2)	29.3995 ± 0.0036	0.77985 ± 0.00021	74.67 ± 0.06	10.906 (0.888)
G 9 / 1.5 + L(5, 0.25)	29.3819 ± 0.0038	0.77510 ± 0.00021	74.97 ± 0.07	11.002 (0.904)
G 9 / 1.5 + L(5, 0.5)	29.2799 ± 0.0044	0.75454 ± 0.00023	76.75 ± 0.08	11.115 (0.887)
G 9 / 1.5 + L(7, 0.1)	29.4742 ± 0.0035	0.79435 ± 0.00017	73.39 ± 0.06	12.570 (1.028)
G 9 / 1.5 + L(7, 0.2)	29.4254 ± 0.0037	0.78492 ± 0.00020	74.22 ± 0.06	12.347 (0.894)
G 9 / 1.5 + L(7, 0.25)	29.4026 ± 0.0039	0.78010 ± 0.00020	74.61 ± 0.07	12.574 (1.059)
G 9 / 1.5 + L(7, 0.5)	29.3287 ± 0.0042	0.76324 ± 0.00021	75.89 ± 0.07	12.733 (0.974)
G 9 / 1.6	29.4080 ± 0.0027	0.80769 ± 0.00014	74.52 ± 0.05	2.987 (0.908)
G 11 / 0.6	24.8300 ± 0.0008	0.40379 ± 0.00014	213.84 ± 0.04	3.489 (0.793)
G 11 / 0.8	27.2607 ± 0.0022	0.55770 ± 0.00027	122.18 ± 0.06	3.506 (0.969)
G 11 / 1.0	28.5066 ± 0.0035	0.66028 ± 0.00029	91.71 ± 0.07	3.472 (1.116)
G 11 / 1.2	29.1329 ± 0.0042	0.72885 ± 0.00025	79.40 ± 0.08	3.535 (0.709)
G 11 / 1.4	29.4005 ± 0.0033	0.77957 ± 0.00021	74.65 ± 0.06	3.620 (0.972)
G 11 / 1.5	29.4227 ± 0.0033	0.79417 ± 0.00018	74.27 ± 0.06	3.505 (0.779)
G 11 / 1.6	29.4034 ± 0.0029	0.80940 ± 0.00013	74.60 ± 0.05	3.388 (0.537)
Med 3	26.0336 ± 0.0042	0.48149 ± 0.00036	162.08 ± 0.16	0.329 (0.013)
Med 5	28.6189 ± 0.0082	0.68707 ± 0.00064	89.37 ± 0.17	3.326 (0.382)
Med 7	29.0048 ± 0.0118	0.77319 ± 0.00042	81.77 ± 0.22	60.777 (3.619)
Med 7 + L(3, 0.1)	28.8446 ± 0.0126	0.74930 ± 0.00053	84.84 ± 0.25	72.828 (1.690)
Med 7 + L(3, 0.2)	28.6397 ± 0.0131	0.72264 ± 0.00060	88.94 ± 0.27	73.172 (3.769)
Med 7 + L(3, 0.25)	28.5244 ± 0.0134	0.70895 ± 0.00065	91.34 ± 0.28	72.820 (2.213)
Med 7 + L(3, 0.5)	27.8400 ± 0.0153	0.63868 ± 0.00082	106.92 ± 0.38	72.056 (1.525)
Med 7 + L(5, 0.1)	29.0191 ± 0.0122	0.77092 ± 0.00045	81.50 ± 0.23	74.758 (3.483)
Med 7 + L(5, 0.2)	28.9498 ± 0.0131	0.75864 ± 0.00048	82.81 ± 0.25	74.136 (2.232)
Med 7 + L(5, 0.25)	28.9219 ± 0.0130	0.75359 ± 0.00047	83.35 ± 0.25	74.106 (2.484)
Med 7 + L(5, 0.5)	28.7800 ± 0.0134	0.73161 ± 0.00052	86.12 ± 0.27	74.495 (1.983)
Med 7 + L(7, 0.1)	29.0387 ± 0.0120	0.77334 ± 0.00042	81.14 ± 0.22	74.904 (2.908)
Med 7 + L(7, 0.2)	29.0238 ± 0.0123	0.76814 ± 0.00046	81.42 ± 0.23	74.431 (2.447)
Med 7 + L(7, 0.25)	29.0040 ± 0.0127	0.76426 ± 0.00048	81.79 ± 0.24	75.825 (3.774)
Med 7 + L(7, 0.5)	28.9339 ± 0.0128	0.74975 ± 0.00045	83.12 ± 0.25	73.999 (2.734)
Med 9	28.5835 ± 0.0104	0.80341 ± 0.00016	90.10 ± 0.22	63.698 (1.123)
Med 11	27.9501 ± 0.0065	0.81202 ± 0.00022	104.25 ± 0.16	68.938 (1.262)
L(3, 0.25)	15.6574 ± 0.0017	0.11104 ± 0.00011	1767.40 ± 0.68	7.143 (1.572)
L(3, 0.5)	13.4347 ± 0.0011	0.07923 ± 0.00011	2948.58 ± 0.74	6.869 (1.226)
L(5, 0.25)	18.2171 ± 0.0031	0.16301 ± 0.00010	980.33 ± 0.69	9.454 (1.288)
L(5, 0.5)	17.5049 ± 0.0026	0.14967 ± 0.00008	1155.02 ± 0.69	8.940 (1.078)

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Configuration	PSNR (dB)	SSIM	MSE	Time (ms)
L(7, 0.25)	18.6441 ± 0.0033	0.17270 ± 0.00010	888.52 ± 0.68	10.632 (1.065)
L(7, 0.5)	18.3239 ± 0.0032	0.16734 ± 0.00009	956.51 ± 0.70	10.318 (1.156)

B.4 Salt-and-pepper noise: 1% density

Table 16: Complete final-grid results: Salt-and-pepper noise, $p = 1\%$.

Configuration	PSNR (dB)	SSIM	MSE	Time (ms)
Noisy input	25.0383 ± 0.0313	0.74063 ± 0.00112	203.82 ± 1.47	N/A
M 3	32.1472 ± 0.0257	0.85450 ± 0.00016	39.66 ± 0.24	0.417 (0.038)
M 3 + L(3, 0.1)	31.7621 ± 0.0276	0.83996 ± 0.00023	43.34 ± 0.28	7.488 (1.046)
M 3 + L(3, 0.2)	31.2854 ± 0.0278	0.82503 ± 0.00033	48.37 ± 0.31	7.900 (1.244)
M 3 + L(3, 0.25)	31.0174 ± 0.0277	0.81708 ± 0.00037	51.45 ± 0.33	7.412 (0.861)
M 3 + L(3, 0.5)	29.5554 ± 0.0275	0.77705 ± 0.00053	72.04 ± 0.46	8.279 (1.017)
M 3 + L(5, 0.1)	32.0506 ± 0.0267	0.85002 ± 0.00019	40.55 ± 0.25	9.503 (0.660)
M 3 + L(5, 0.2)	31.9291 ± 0.0276	0.84502 ± 0.00023	41.70 ± 0.27	9.780 (1.348)
M 3 + L(5, 0.25)	31.8579 ± 0.0285	0.84250 ± 0.00023	42.39 ± 0.28	9.460 (1.251)
M 3 + L(5, 0.5)	31.3644 ± 0.0291	0.82900 ± 0.00033	47.50 ± 0.32	10.014 (1.988)
M 3 + L(7, 0.1)	32.0917 ± 0.0261	0.85182 ± 0.00020	40.17 ± 0.24	11.589 (1.815)
M 3 + L(7, 0.2)	32.0217 ± 0.0275	0.84839 ± 0.00021	40.82 ± 0.26	10.881 (1.076)
M 3 + L(7, 0.25)	31.9815 ± 0.0282	0.84680 ± 0.00021	41.20 ± 0.27	10.927 (0.855)
M 3 + L(7, 0.5)	31.6863 ± 0.0295	0.83784 ± 0.00028	44.10 ± 0.30	11.178 (2.685)
M 5	31.0767 ± 0.0131	0.87175 ± 0.00011	50.75 ± 0.15	0.557 (0.121)
M 7	29.5087 ± 0.0091	0.86079 ± 0.00019	72.81 ± 0.15	1.834 (0.396)
M 9	28.2392 ± 0.0062	0.84383 ± 0.00026	97.54 ± 0.14	1.807 (0.597)
M 11	27.2207 ± 0.0039	0.82811 ± 0.00014	123.31 ± 0.11	1.919 (0.397)
G 3 / 0.6	30.6295 ± 0.0292	0.83013 ± 0.00056	56.25 ± 0.38	0.702 (0.189)
G 3 / 0.8	32.0301 ± 0.0289	0.85344 ± 0.00031	40.75 ± 0.27	0.682 (0.285)
G 3 / 1.0	32.3065 ± 0.0281	0.85872 ± 0.00023	38.23 ± 0.25	0.735 (0.203)
G 3 / 1.2	32.3335 ± 0.0275	0.85913 ± 0.00019	38.00 ± 0.24	0.759 (0.270)
G 3 / 1.4	32.3131 ± 0.0270	0.85875 ± 0.00017	38.17 ± 0.24	0.661 (0.116)
G 3 / 1.5	32.3011 ± 0.0268	0.85851 ± 0.00018	38.28 ± 0.24	0.738 (0.204)
G 3 / 1.6	32.2826 ± 0.0269	0.85794 ± 0.00017	38.44 ± 0.24	0.733 (0.229)
G 5 / 0.6	30.6553 ± 0.0296	0.83082 ± 0.00055	55.92 ± 0.38	1.379 (0.678)
G 5 / 0.8	32.3358 ± 0.0278	0.86225 ± 0.00025	37.98 ± 0.24	1.289 (0.523)
G 5 / 1.0	32.6387 ± 0.0243	0.87682 ± 0.00011	35.42 ± 0.20	1.149 (0.470)
G 5 / 1.0 + L(3, 0.1)	32.5927 ± 0.0260	0.87085 ± 0.00013	35.79 ± 0.21	8.258 (1.116)
G 5 / 1.0 + L(3, 0.2)	32.4758 ± 0.0277	0.86431 ± 0.00020	36.77 ± 0.23	9.540 (13.416)
G 5 / 1.0 + L(3, 0.25)	32.3859 ± 0.0279	0.86084 ± 0.00023	37.54 ± 0.24	9.360 (6.636)
G 5 / 1.0 + L(3, 0.5)	31.7002 ± 0.0298	0.84195 ± 0.00034	43.96 ± 0.30	7.955 (0.880)
G 5 / 1.0 + L(5, 0.1)	32.6194 ± 0.0256	0.87301 ± 0.00013	35.57 ± 0.21	10.204 (1.183)
G 5 / 1.0 + L(5, 0.2)	32.5763 ± 0.0265	0.86891 ± 0.00017	35.93 ± 0.22	9.913 (0.806)
G 5 / 1.0 + L(5, 0.25)	32.5436 ± 0.0271	0.86674 ± 0.00018	36.20 ± 0.23	9.679 (0.799)
G 5 / 1.0 + L(5, 0.5)	32.2350 ± 0.0294	0.85534 ± 0.00025	38.87 ± 0.26	10.023 (0.619)
G 5 / 1.0 + L(7, 0.1)	32.6353 ± 0.0254	0.87436 ± 0.00013	35.45 ± 0.21	11.294 (0.505)
G 5 / 1.0 + L(7, 0.2)	32.6160 ± 0.0266	0.87144 ± 0.00014	35.60 ± 0.22	11.899 (1.869)
G 5 / 1.0 + L(7, 0.25)	32.6066 ± 0.0270	0.87012 ± 0.00014	35.68 ± 0.22	11.903 (1.358)
G 5 / 1.0 + L(7, 0.5)	32.4365 ± 0.0290	0.86190 ± 0.00018	37.11 ± 0.25	11.821 (1.371)

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Configuration	PSNR (dB)	SSIM	MSE	Time (ms)
G 5 / 1.2	32.4897 ± 0.0206	0.88373 ± 0.00010	36.65 ± 0.17	1.614 (0.695)
G 5 / 1.4	32.2536 ± 0.0183	0.88564 ± 0.00011	38.70 ± 0.16	1.481 (0.674)
G 5 / 1.5	32.1210 ± 0.0173	0.88574 ± 0.00012	39.90 ± 0.16	1.298 (0.547)
G 5 / 1.6	32.0350 ± 0.0168	0.88558 ± 0.00013	40.70 ± 0.16	1.510 (0.647)
G 7 / 0.6	30.6553 ± 0.0296	0.83082 ± 0.00055	55.92 ± 0.38	2.723 (0.516)
G 7 / 0.8	32.3358 ± 0.0278	0.86225 ± 0.00025	37.98 ± 0.24	2.558 (0.898)
G 7 / 1.0	32.6288 ± 0.0236	0.87818 ± 0.00009	35.50 ± 0.19	2.650 (0.445)
G 7 / 1.2	32.3797 ± 0.0192	0.88670 ± 0.00010	37.59 ± 0.17	2.717 (0.721)
G 7 / 1.4	31.9621 ± 0.0162	0.88983 ± 0.00014	41.39 ± 0.15	2.616 (0.526)
G 7 / 1.5	31.7495 ± 0.0147	0.89021 ± 0.00015	43.46 ± 0.15	2.506 (0.954)
G 7 / 1.6	31.5581 ± 0.0138	0.88990 ± 0.00016	45.42 ± 0.14	2.650 (0.788)
G 9 / 0.6	30.6553 ± 0.0296	0.83082 ± 0.00055	55.92 ± 0.38	2.914 (0.936)
G 9 / 0.8	32.3358 ± 0.0278	0.86225 ± 0.00025	37.98 ± 0.24	3.210 (0.969)
G 9 / 1.0	32.6288 ± 0.0236	0.87818 ± 0.00009	35.50 ± 0.19	3.151 (0.984)
G 9 / 1.2	32.3797 ± 0.0192	0.88670 ± 0.00010	37.59 ± 0.17	3.141 (1.341)
G 9 / 1.4	31.8859 ± 0.0155	0.89047 ± 0.00014	42.12 ± 0.15	3.229 (0.957)
G 9 / 1.5	31.6277 ± 0.0140	0.89043 ± 0.00015	44.70 ± 0.14	3.273 (1.139)
G 9 / 1.6	31.3634 ± 0.0131	0.89004 ± 0.00016	47.51 ± 0.14	3.318 (0.828)
G 11 / 0.6	30.6553 ± 0.0296	0.83082 ± 0.00055	55.92 ± 0.38	3.760 (1.444)
G 11 / 0.8	32.3358 ± 0.0278	0.86225 ± 0.00025	37.98 ± 0.24	3.668 (1.083)
G 11 / 1.0	32.6288 ± 0.0236	0.87818 ± 0.00009	35.50 ± 0.19	3.399 (1.284)
G 11 / 1.2	32.3797 ± 0.0192	0.88670 ± 0.00010	37.59 ± 0.17	3.605 (1.323)
G 11 / 1.4	31.8601 ± 0.0152	0.89024 ± 0.00014	42.37 ± 0.15	3.600 (1.305)
G 11 / 1.5	31.6277 ± 0.0140	0.89043 ± 0.00015	44.70 ± 0.14	3.443 (1.285)
G 11 / 1.6	31.3289 ± 0.0128	0.88994 ± 0.00015	47.88 ± 0.14	3.970 (1.858)
Med 3	38.2479 ± 0.0093	0.96514 ± 0.00002	9.73 ± 0.02	0.324 (0.024)
Med 3 + L(3, 0.1)	38.5215 ± 0.0065	0.96654 ± 0.00002	9.14 ± 0.01	8.606 (1.619)
Med 3 + L(3, 0.2)	38.2450 ± 0.0030	0.96605 ± 0.00003	9.74 ± 0.01	7.646 (1.299)
Med 3 + L(3, 0.25)	37.9206 ± 0.0010	0.96492 ± 0.00003	10.50 ± 0.00	7.187 (0.709)
Med 3 + L(3, 0.5)	35.5112 ± 0.0058	0.95469 ± 0.00004	18.28 ± 0.02	7.992 (1.143)
Med 3 + L(5, 0.1)	38.5616 ± 0.0082	0.96630 ± 0.00002	9.06 ± 0.02	9.348 (1.054)
Med 3 + L(5, 0.2)	38.7452 ± 0.0058	0.96752 ± 0.00003	8.68 ± 0.01	9.229 (0.571)
Med 3 + L(5, 0.25)	38.7671 ± 0.0048	0.96794 ± 0.00002	8.64 ± 0.01	9.217 (0.772)
Med 3 + L(5, 0.5)	38.2218 ± 0.0031	0.96836 ± 0.00003	9.79 ± 0.01	9.420 (0.968)
Med 3 + L(7, 0.1)	38.4628 ± 0.0081	0.96582 ± 0.00002	9.26 ± 0.02	11.310 (2.282)
Med 3 + L(7, 0.2)	38.6178 ± 0.0069	0.96670 ± 0.00002	8.94 ± 0.01	11.126 (2.134)
Med 3 + L(7, 0.25)	38.6586 ± 0.0060	0.96706 ± 0.00002	8.86 ± 0.01	10.746 (1.162)
Med 3 + L(7, 0.5)	38.4680 ± 0.0015	0.96787 ± 0.00003	9.25 ± 0.00	11.450 (1.057)
Med 5	34.3145 ± 0.0116	0.93445 ± 0.00002	24.08 ± 0.06	3.491 (0.732)
Med 7	32.2140 ± 0.0095	0.91086 ± 0.00001	39.06 ± 0.09	50.886 (0.374)
Med 9	30.7121 ± 0.0071	0.89230 ± 0.00001	55.19 ± 0.09	55.329 (0.717)
Med 11	29.5367 ± 0.0067	0.87718 ± 0.00003	72.35 ± 0.11	60.836 (0.827)
L(3, 0.25)	24.6682 ± 0.0298	0.72682 ± 0.00111	221.96 ± 1.52	7.119 (0.750)
L(3, 0.5)	23.8478 ± 0.0230	0.69181 ± 0.00117	268.10 ± 1.42	7.405 (1.150)
L(5, 0.25)	24.9705 ± 0.0310	0.73905 ± 0.00112	207.03 ± 1.48	8.965 (0.580)
L(5, 0.5)	24.7798 ± 0.0299	0.73226 ± 0.00110	216.32 ± 1.49	9.014 (0.905)
L(7, 0.25)	25.0084 ± 0.0312	0.74036 ± 0.00112	205.23 ± 1.48	10.543 (0.945)
L(7, 0.5)	24.9236 ± 0.0308	0.73781 ± 0.00111	209.28 ± 1.49	10.134 (0.813)

B.5 Salt-and-pepper noise: 3% density

Table 17: Complete final-grid results: Salt-and-pepper noise, $p = 3\%$.

Configuration	PSNR (dB)	SSIM	MSE	Time (ms)
Noisy input	20.2892 ± 0.0058	0.44027 ± 0.00076	608.35 ± 0.81	N/A
M 3	28.6612 ± 0.0119	0.70030 ± 0.00049	88.50 ± 0.24	0.511 (0.102)
M 5	29.5823 ± 0.0088	0.80084 ± 0.00046	71.59 ± 0.14	0.628 (0.908)
M 5 + L(3, 0.1)	29.4218 ± 0.0080	0.78034 ± 0.00048	74.28 ± 0.14	8.782 (2.414)
M 5 + L(3, 0.2)	29.2296 ± 0.0073	0.75820 ± 0.00052	77.65 ± 0.13	9.239 (3.856)
M 5 + L(3, 0.25)	29.1223 ± 0.0068	0.74700 ± 0.00053	79.59 ± 0.12	9.148 (3.335)
M 5 + L(3, 0.5)	28.4610 ± 0.0038	0.68763 ± 0.00066	92.68 ± 0.08	8.697 (2.978)
M 5 + L(5, 0.1)	29.5986 ± 0.0086	0.79901 ± 0.00045	71.32 ± 0.14	11.668 (2.603)
M 5 + L(5, 0.2)	29.5375 ± 0.0082	0.78803 ± 0.00052	72.33 ± 0.14	10.118 (2.622)
M 5 + L(5, 0.25)	29.5108 ± 0.0080	0.78339 ± 0.00051	72.78 ± 0.13	11.373 (2.498)
M 5 + L(5, 0.5)	29.3840 ± 0.0079	0.76445 ± 0.00057	74.93 ± 0.14	10.984 (5.697)
M 5 + L(7, 0.1)	29.6246 ± 0.0088	0.80105 ± 0.00044	70.90 ± 0.14	12.969 (3.862)
M 5 + L(7, 0.2)	29.6163 ± 0.0091	0.79661 ± 0.00051	71.03 ± 0.15	12.917 (5.093)
M 5 + L(7, 0.25)	29.6021 ± 0.0090	0.79315 ± 0.00055	71.26 ± 0.15	12.587 (2.533)
M 5 + L(7, 0.5)	29.5496 ± 0.0087	0.77994 ± 0.00056	72.13 ± 0.15	14.205 (4.585)
M 7	28.7707 ± 0.0066	0.82739 ± 0.00018	86.30 ± 0.13	2.156 (0.426)
M 9	27.7862 ± 0.0046	0.82626 ± 0.00023	108.26 ± 0.11	2.380 (0.741)
M 11	26.8964 ± 0.0017	0.81786 ± 0.00024	132.87 ± 0.05	2.472 (0.907)
G 3 / 0.6	26.0989 ± 0.0102	0.61445 ± 0.00062	159.66 ± 0.38	1.228 (0.487)
G 3 / 0.8	27.9087 ± 0.0121	0.67592 ± 0.00053	105.25 ± 0.29	0.848 (0.336)
G 3 / 1.0	28.4451 ± 0.0124	0.69541 ± 0.00049	93.02 ± 0.27	0.847 (0.497)
G 3 / 1.2	28.6127 ± 0.0126	0.70121 ± 0.00050	89.50 ± 0.26	0.829 (0.397)
G 3 / 1.4	28.6688 ± 0.0125	0.70294 ± 0.00049	88.35 ± 0.25	0.945 (0.480)
G 3 / 1.5	28.6804 ± 0.0122	0.70322 ± 0.00048	88.11 ± 0.25	1.330 (0.577)
G 3 / 1.6	28.6862 ± 0.0123	0.70311 ± 0.00049	88.00 ± 0.25	0.959 (0.656)
G 5 / 0.6	26.1351 ± 0.0102	0.61591 ± 0.00062	158.33 ± 0.37	1.509 (0.718)
G 5 / 0.8	28.4664 ± 0.0129	0.70005 ± 0.00052	92.56 ± 0.27	1.409 (0.874)
G 5 / 1.0	29.4806 ± 0.0140	0.74961 ± 0.00052	73.29 ± 0.24	1.685 (0.693)
G 5 / 1.2	29.9099 ± 0.0142	0.78094 ± 0.00055	66.39 ± 0.22	1.422 (0.636)
G 5 / 1.4	30.0253 ± 0.0139	0.79670 ± 0.00055	64.65 ± 0.21	1.796 (0.256)
G 5 / 1.5	30.0356 ± 0.0133	0.80210 ± 0.00055	64.50 ± 0.20	1.298 (0.505)
G 5 / 1.6	30.0305 ± 0.0132	0.80483 ± 0.00056	64.57 ± 0.20	1.596 (1.190)
G 7 / 0.6	26.1351 ± 0.0102	0.61591 ± 0.00062	158.33 ± 0.37	2.719 (1.308)
G 7 / 0.8	28.4664 ± 0.0129	0.70005 ± 0.00052	92.56 ± 0.27	2.600 (1.014)
G 7 / 1.0	29.5590 ± 0.0141	0.75461 ± 0.00052	71.97 ± 0.23	2.706 (1.474)
G 7 / 1.2	30.0144 ± 0.0141	0.79273 ± 0.00055	64.81 ± 0.21	2.880 (1.674)
G 7 / 1.4	30.1131 ± 0.0131	0.81756 ± 0.00053	63.35 ± 0.19	2.725 (1.215)
G 7 / 1.4 + L(3, 0.1)	30.0938 ± 0.0135	0.81074 ± 0.00055	63.64 ± 0.20	10.663 (2.658)
G 7 / 1.4 + L(3, 0.2)	30.0634 ± 0.0137	0.80165 ± 0.00056	64.08 ± 0.20	10.992 (16.703)
G 7 / 1.4 + L(3, 0.25)	30.0423 ± 0.0137	0.79722 ± 0.00057	64.39 ± 0.20	10.696 (1.757)
G 7 / 1.4 + L(3, 0.5)	29.8398 ± 0.0137	0.77408 ± 0.00059	67.47 ± 0.21	10.716 (3.017)
G 7 / 1.4 + L(5, 0.1)	30.1179 ± 0.0134	0.81386 ± 0.00054	63.28 ± 0.20	12.640 (1.816)
G 7 / 1.4 + L(5, 0.2)	30.0858 ± 0.0137	0.80619 ± 0.00055	63.75 ± 0.20	12.549 (2.054)
G 7 / 1.4 + L(5, 0.25)	30.0763 ± 0.0137	0.80303 ± 0.00056	63.89 ± 0.20	12.779 (2.278)
G 7 / 1.4 + L(5, 0.5)	29.9618 ± 0.0138	0.78612 ± 0.00059	65.60 ± 0.21	13.153 (2.611)
G 7 / 1.4 + L(7, 0.1)	30.1426 ± 0.0133	0.81639 ± 0.00053	62.92 ± 0.19	13.605 (1.951)

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Configuration	PSNR (dB)	SSIM	MSE	Time (ms)
G 7 / 1.4 + L(7, 0.2)	30.0996 ± 0.0138	0.80905 ± 0.00057	63.55 ± 0.20	13.893 (1.616)
G 7 / 1.4 + L(7, 0.25)	30.0930 ± 0.0138	0.80641 ± 0.00055	63.65 ± 0.20	14.013 (2.315)
G 7 / 1.4 + L(7, 0.5)	30.0235 ± 0.0133	0.79330 ± 0.00061	64.67 ± 0.20	13.425 (1.486)
G 7 / 1.5	30.0903 ± 0.0126	0.82609 ± 0.00051	63.69 ± 0.19	3.066 (0.860)
G 7 / 1.6	30.0435 ± 0.0118	0.83201 ± 0.00049	64.38 ± 0.17	2.734 (1.463)
G 9 / 0.6	26.1351 ± 0.0102	0.61591 ± 0.00062	158.33 ± 0.37	3.266 (1.509)
G 9 / 0.8	28.4664 ± 0.0129	0.70005 ± 0.00052	92.56 ± 0.27	3.005 (1.283)
G 9 / 1.0	29.5590 ± 0.0141	0.75461 ± 0.00052	71.97 ± 0.23	3.169 (1.190)
G 9 / 1.2	30.0144 ± 0.0141	0.79273 ± 0.00055	64.81 ± 0.21	3.362 (1.476)
G 9 / 1.4	30.1090 ± 0.0126	0.82103 ± 0.00050	63.41 ± 0.18	3.650 (1.388)
G 9 / 1.5	30.0614 ± 0.0123	0.82999 ± 0.00049	64.11 ± 0.18	2.976 (1.091)
G 9 / 1.6	29.9814 ± 0.0116	0.83772 ± 0.00046	65.30 ± 0.17	3.320 (1.145)
G 11 / 0.6	26.1351 ± 0.0102	0.61591 ± 0.00062	158.33 ± 0.37	3.886 (1.319)
G 11 / 0.8	28.4664 ± 0.0129	0.70005 ± 0.00052	92.56 ± 0.27	2.999 (0.488)
G 11 / 1.0	29.5590 ± 0.0141	0.75461 ± 0.00052	71.97 ± 0.23	3.445 (1.295)
G 11 / 1.2	30.0144 ± 0.0141	0.79273 ± 0.00055	64.81 ± 0.21	3.382 (1.175)
G 11 / 1.4	30.1028 ± 0.0127	0.82160 ± 0.00051	63.50 ± 0.19	3.139 (0.724)
G 11 / 1.5	30.0614 ± 0.0123	0.82999 ± 0.00049	64.11 ± 0.18	3.311 (1.166)
G 11 / 1.6	29.9693 ± 0.0113	0.83866 ± 0.00046	65.49 ± 0.17	3.120 (0.484)
Med 3	37.8470 ± 0.0330	0.96393 ± 0.00006	10.68 ± 0.08	0.449 (0.347)
Med 3 + L(3, 0.1)	37.9997 ± 0.0465	0.96507 ± 0.00007	10.31 ± 0.11	9.498 (1.766)
Med 3 + L(3, 0.2)	37.6488 ± 0.0545	0.96418 ± 0.00008	11.17 ± 0.14	10.407 (5.648)
Med 3 + L(3, 0.25)	37.3145 ± 0.0569	0.96286 ± 0.00009	12.07 ± 0.16	10.479 (2.242)
Med 3 + L(3, 0.5)	34.9908 ± 0.0530	0.95171 ± 0.00013	20.61 ± 0.25	9.744 (10.778)
Med 3 + L(5, 0.1)	38.1057 ± 0.0387	0.96505 ± 0.00006	10.06 ± 0.09	12.165 (2.613)
Med 3 + L(5, 0.2)	38.2394 ± 0.0440	0.96618 ± 0.00006	9.75 ± 0.10	13.619 (4.835)
Med 3 + L(5, 0.25)	38.2426 ± 0.0459	0.96655 ± 0.00006	9.75 ± 0.10	12.210 (4.236)
Med 3 + L(5, 0.5)	37.6842 ± 0.0497	0.96673 ± 0.00007	11.08 ± 0.13	11.995 (2.073)
Med 3 + L(7, 0.1)	38.0289 ± 0.0368	0.96459 ± 0.00006	10.24 ± 0.09	13.941 (5.851)
Med 3 + L(7, 0.2)	38.1543 ± 0.0397	0.96544 ± 0.00006	9.95 ± 0.09	14.119 (2.653)
Med 3 + L(7, 0.25)	38.1819 ± 0.0408	0.96577 ± 0.00006	9.88 ± 0.09	15.982 (3.264)
Med 3 + L(7, 0.5)	37.9698 ± 0.0446	0.96646 ± 0.00006	10.38 ± 0.11	12.909 (2.607)
Med 5	34.1864 ± 0.0039	0.93375 ± 0.00005	24.80 ± 0.02	3.795 (1.146)
Med 7	32.1496 ± 0.0086	0.91032 ± 0.00004	39.64 ± 0.08	56.708 (3.746)
Med 9	30.6691 ± 0.0070	0.89187 ± 0.00004	55.74 ± 0.09	64.116 (13.034)
Med 11	29.5102 ± 0.0040	0.87688 ± 0.00002	72.79 ± 0.07	70.589 (7.593)
L(3, 0.25)	20.0186 ± 0.0059	0.43021 ± 0.00071	647.47 ± 0.88	9.321 (2.509)
L(3, 0.5)	19.4159 ± 0.0050	0.40049 ± 0.00064	743.85 ± 0.85	8.851 (1.383)
L(5, 0.25)	20.2479 ± 0.0057	0.44038 ± 0.00074	614.18 ± 0.81	10.986 (1.553)
L(5, 0.5)	20.1294 ± 0.0054	0.43562 ± 0.00072	631.16 ± 0.79	12.085 (8.571)
L(7, 0.25)	20.2732 ± 0.0058	0.44125 ± 0.00075	610.60 ± 0.81	13.974 (14.226)
L(7, 0.5)	20.2265 ± 0.0057	0.44017 ± 0.00073	617.20 ± 0.81	13.291 (2.531)

B.6 Salt-and-pepper noise: 5% density

Table 18: Complete final-grid results: Salt-and-pepper noise, $p = 5\%$.

Configuration	PSNR (dB)	SSIM	MSE	Time (ms)
Noisy input	18.0649 ± 0.0108	0.28699 ± 0.00171	1015.29 ± 2.52	N/A
M 3	26.6668 ± 0.0066	0.58972 ± 0.00126	140.09 ± 0.21	0.608 (0.212)
M 5	28.3379 ± 0.0179	0.74163 ± 0.00012	95.34 ± 0.39	0.683 (0.194)
M 5 + L(3, 0.1)	28.1114 ± 0.0174	0.71269 ± 0.00028	100.45 ± 0.40	9.625 (2.643)
M 5 + L(3, 0.2)	27.8475 ± 0.0160	0.68226 ± 0.00041	106.74 ± 0.39	10.459 (1.592)
M 5 + L(3, 0.25)	27.7053 ± 0.0159	0.66706 ± 0.00050	110.29 ± 0.40	9.960 (2.581)
M 5 + L(3, 0.5)	26.8889 ± 0.0135	0.59135 ± 0.00074	133.10 ± 0.42	9.458 (1.698)
M 5 + L(5, 0.1)	28.3074 ± 0.0184	0.73659 ± 0.00014	96.02 ± 0.41	11.399 (2.368)
M 5 + L(5, 0.2)	28.2082 ± 0.0176	0.72158 ± 0.00024	98.23 ± 0.40	11.502 (1.633)
M 5 + L(5, 0.25)	28.1669 ± 0.0180	0.71587 ± 0.00024	99.17 ± 0.41	12.176 (3.402)
M 5 + L(5, 0.5)	27.9464 ± 0.0176	0.68963 ± 0.00033	104.34 ± 0.42	11.531 (2.244)
M 5 + L(7, 0.1)	28.3536 ± 0.0188	0.74109 ± 0.00011	95.00 ± 0.41	12.475 (1.316)
M 5 + L(7, 0.2)	28.2943 ± 0.0185	0.73228 ± 0.00017	96.31 ± 0.41	13.208 (2.317)
M 5 + L(7, 0.25)	28.2672 ± 0.0187	0.72765 ± 0.00016	96.91 ± 0.42	13.348 (2.693)
M 5 + L(7, 0.5)	28.1477 ± 0.0186	0.71048 ± 0.00023	99.61 ± 0.43	13.081 (1.975)
M 7	28.0009 ± 0.0112	0.79669 ± 0.00038	103.04 ± 0.27	2.097 (1.469)
M 9	27.2487 ± 0.0036	0.80908 ± 0.00022	122.52 ± 0.10	2.416 (0.815)
M 11	26.4843 ± 0.0022	0.80749 ± 0.00011	146.10 ± 0.08	2.388 (0.886)
G 3 / 0.6	23.8878 ± 0.0092	0.47509 ± 0.00190	265.64 ± 0.56	0.905 (0.588)
G 3 / 0.8	25.7660 ± 0.0056	0.55272 ± 0.00166	172.38 ± 0.22	0.920 (0.286)
G 3 / 1.0	26.3565 ± 0.0043	0.57953 ± 0.00149	150.46 ± 0.15	1.184 (1.510)
G 3 / 1.2	26.5570 ± 0.0046	0.58831 ± 0.00139	143.68 ± 0.15	0.925 (1.036)
G 3 / 1.4	26.6319 ± 0.0049	0.59125 ± 0.00133	141.22 ± 0.16	0.832 (0.732)
G 3 / 1.5	26.6495 ± 0.0051	0.59183 ± 0.00131	140.65 ± 0.17	0.766 (0.252)
G 3 / 1.6	26.6619 ± 0.0053	0.59201 ± 0.00129	140.25 ± 0.17	1.240 (0.628)
G 5 / 0.6	23.9258 ± 0.0092	0.47673 ± 0.00191	263.33 ± 0.56	2.046 (0.499)
G 5 / 0.8	26.3762 ± 0.0049	0.58390 ± 0.00154	149.78 ± 0.17	1.520 (1.123)
G 5 / 1.0	27.5674 ± 0.0079	0.65295 ± 0.00108	113.85 ± 0.21	1.727 (1.050)
G 5 / 1.2	28.1802 ± 0.0124	0.69978 ± 0.00067	98.87 ± 0.28	1.722 (0.982)
G 5 / 1.4	28.4284 ± 0.0150	0.72495 ± 0.00044	93.38 ± 0.32	1.634 (1.018)
G 5 / 1.5	28.4983 ± 0.0161	0.73406 ± 0.00034	91.89 ± 0.34	1.963 (1.452)
G 5 / 1.6	28.5287 ± 0.0168	0.73884 ± 0.00029	91.25 ± 0.35	2.004 (0.818)
G 7 / 0.6	23.9258 ± 0.0092	0.47673 ± 0.00191	263.33 ± 0.56	2.884 (0.639)
G 7 / 0.8	26.3762 ± 0.0049	0.58390 ± 0.00154	149.78 ± 0.17	3.167 (0.872)
G 7 / 1.0	27.6716 ± 0.0085	0.66009 ± 0.00102	111.15 ± 0.22	2.554 (0.927)
G 7 / 1.2	28.3639 ± 0.0140	0.71717 ± 0.00055	94.78 ± 0.31	2.676 (0.924)
G 7 / 1.4	28.6792 ± 0.0176	0.75729 ± 0.00019	88.14 ± 0.36	3.297 (1.425)
G 7 / 1.5	28.7483 ± 0.0185	0.77183 ± 0.00007	86.75 ± 0.37	2.868 (0.633)
G 7 / 1.6	28.7762 ± 0.0192	0.78246 ± 0.00007	86.19 ± 0.38	3.210 (0.911)
G 9 / 0.6	23.9258 ± 0.0092	0.47673 ± 0.00191	263.33 ± 0.56	3.413 (0.792)
G 9 / 0.8	26.3762 ± 0.0049	0.58390 ± 0.00154	149.78 ± 0.17	3.399 (0.848)
G 9 / 1.0	27.6716 ± 0.0085	0.66009 ± 0.00102	111.15 ± 0.22	3.019 (0.845)
G 9 / 1.2	28.3639 ± 0.0140	0.71717 ± 0.00055	94.78 ± 0.31	3.720 (1.454)
G 9 / 1.4	28.7093 ± 0.0177	0.76272 ± 0.00016	87.53 ± 0.36	3.112 (1.056)
G 9 / 1.5	28.7667 ± 0.0184	0.77838 ± 0.00003	86.38 ± 0.37	2.691 (0.850)
G 9 / 1.6	28.7862 ± 0.0186	0.79229 ± 0.00009	85.99 ± 0.37	2.619 (0.331)
G 11 / 0.6	23.9258 ± 0.0092	0.47673 ± 0.00191	263.33 ± 0.56	3.395 (1.669)
G 11 / 0.8	26.3762 ± 0.0049	0.58390 ± 0.00154	149.78 ± 0.17	3.307 (1.135)

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Configuration	PSNR (dB)	SSIM	MSE	Time (ms)
G 11 / 1.0	27.6716 ± 0.0085	0.66009 ± 0.00102	111.15 ± 0.22	3.464 (6.579)
G 11 / 1.2	28.3639 ± 0.0140	0.71717 ± 0.00055	94.78 ± 0.31	3.763 (2.320)
G 11 / 1.4	28.7128 ± 0.0177	0.76391 ± 0.00015	87.46 ± 0.36	4.158 (2.062)
G 11 / 1.5	28.7667 ± 0.0184	0.77838 ± 0.00003	86.38 ± 0.37	3.447 (0.731)
G 11 / 1.6	28.7862 ± 0.0185	0.79405 ± 0.00008	85.99 ± 0.37	4.042 (1.001)
G 11 / 1.6 + L(3, 0.1)	28.7883 ± 0.0186	0.78938 ± 0.00005	85.95 ± 0.37	12.266 (4.143)
G 11 / 1.6 + L(3, 0.2)	28.7636 ± 0.0191	0.77909 ± 0.00003	86.44 ± 0.38	13.182 (4.161)
G 11 / 1.6 + L(3, 0.25)	28.7522 ± 0.0188	0.77464 ± 0.00006	86.67 ± 0.38	11.273 (1.885)
G 11 / 1.6 + L(3, 0.5)	28.6469 ± 0.0186	0.75265 ± 0.00019	88.80 ± 0.38	11.377 (1.798)
G 11 / 1.6 + L(5, 0.1)	28.8094 ± 0.0189	0.79253 ± 0.00007	85.54 ± 0.37	13.050 (1.698)
G 11 / 1.6 + L(5, 0.2)	28.7752 ± 0.0189	0.78321 ± 0.00004	86.21 ± 0.38	13.133 (1.386)
G 11 / 1.6 + L(5, 0.25)	28.7675 ± 0.0189	0.77944 ± 0.00004	86.36 ± 0.38	13.535 (1.494)
G 11 / 1.6 + L(5, 0.5)	28.7113 ± 0.0192	0.76297 ± 0.00011	87.49 ± 0.39	13.440 (1.465)
G 11 / 1.6 + L(7, 0.1)	28.8169 ± 0.0189	0.79373 ± 0.00008	85.39 ± 0.37	14.573 (2.220)
G 11 / 1.6 + L(7, 0.2)	28.7931 ± 0.0192	0.78684 ± 0.00006	85.86 ± 0.38	15.940 (2.319)
G 11 / 1.6 + L(7, 0.25)	28.7808 ± 0.0191	0.78293 ± 0.00004	86.10 ± 0.38	17.058 (2.696)
G 11 / 1.6 + L(7, 0.5)	28.7437 ± 0.0196	0.76904 ± 0.00003	86.84 ± 0.39	15.948 (3.919)
Med 3	37.4485 ± 0.0597	0.96265 ± 0.00003	11.70 ± 0.16	0.437 (0.166)
Med 3 + L(3, 0.1)	37.4920 ± 0.0641	0.96350 ± 0.00003	11.59 ± 0.17	9.398 (2.034)
Med 3 + L(3, 0.2)	37.0790 ± 0.0641	0.96221 ± 0.00003	12.74 ± 0.19	9.898 (2.036)
Med 3 + L(3, 0.25)	36.7341 ± 0.0627	0.96068 ± 0.00003	13.79 ± 0.20	11.230 (3.199)
Med 3 + L(3, 0.5)	34.4733 ± 0.0492	0.94852 ± 0.00004	23.22 ± 0.26	10.207 (1.853)
Med 3 + L(5, 0.1)	37.6595 ± 0.0621	0.96371 ± 0.00003	11.15 ± 0.16	13.155 (2.717)
Med 3 + L(5, 0.2)	37.7493 ± 0.0626	0.96474 ± 0.00004	10.92 ± 0.16	12.355 (2.826)
Med 3 + L(5, 0.25)	37.7372 ± 0.0624	0.96507 ± 0.00004	10.95 ± 0.16	13.353 (3.829)
Med 3 + L(5, 0.5)	37.1676 ± 0.0549	0.96500 ± 0.00004	12.48 ± 0.16	12.744 (2.786)
Med 3 + L(7, 0.1)	37.6022 ± 0.0612	0.96329 ± 0.00003	11.30 ± 0.16	13.716 (3.189)
Med 3 + L(7, 0.2)	37.7003 ± 0.0626	0.96409 ± 0.00003	11.04 ± 0.16	12.755 (1.400)
Med 3 + L(7, 0.25)	37.7172 ± 0.0617	0.96440 ± 0.00004	11.00 ± 0.16	13.919 (2.681)
Med 3 + L(7, 0.5)	37.4908 ± 0.0569	0.96497 ± 0.00005	11.59 ± 0.15	14.317 (2.476)
Med 5	34.0706 ± 0.0044	0.93294 ± 0.00003	25.47 ± 0.03	3.515 (0.452)
Med 7	32.0836 ± 0.0071	0.90973 ± 0.00006	40.25 ± 0.07	56.394 (4.395)
Med 9	30.6218 ± 0.0070	0.89139 ± 0.00007	56.35 ± 0.09	62.148 (8.410)
Med 11	29.4752 ± 0.0098	0.87647 ± 0.00011	73.38 ± 0.17	66.984 (8.461)
L(3, 0.25)	17.8166 ± 0.0117	0.28101 ± 0.00163	1075.02 ± 2.89	8.959 (2.250)
L(3, 0.5)	17.2678 ± 0.0136	0.25952 ± 0.00148	1219.85 ± 3.81	10.133 (3.092)
L(5, 0.25)	18.0295 ± 0.0108	0.28818 ± 0.00169	1023.60 ± 2.54	10.676 (1.720)
L(5, 0.5)	17.9272 ± 0.0109	0.28545 ± 0.00165	1048.01 ± 2.62	11.523 (2.056)
L(7, 0.25)	18.0522 ± 0.0108	0.28855 ± 0.00170	1018.28 ± 2.52	12.416 (1.702)
L(7, 0.5)	18.0138 ± 0.0107	0.28849 ± 0.00169	1027.31 ± 2.54	13.442 (2.505)